American Terawatt

HVDC DESIGN

HVDC Design

Design notes for VSC HVDC transmission

Revision history

REV.	DATE	AUTHOR	DESCRIPTION
0.1	Oct. 19, 2026	Eugen Starschich	Initial document: creepage distance of DC OHL insulators
0.2	Oct. 20, 2026	Eugen Starschich	Added AC power transmission: line parameters, two-port and distributed representation
0.3	Oct. 20, 2026	Eugen Starschich	Split the AC material into the general representation, overhead lines, cables and the worked example
0.4	Oct. 20, 2026	Eugen Starschich	Added natural power, transmission limits, angular stability and series compensation
0.5	Oct. 20, 2026	Eugen Starschich	All AC material collected under one chapter; table captions below the table; every figure referenced from the text
0.6	Oct. 20, 2026	Eugen Starschich	Added the general two-port chapter; transferred the remaining presentation drawings and the symbols list
0.7	Oct. 20, 2026	Eugen Starschich	Two-port theory moved to Appendix A as background material
0.8	Oct. 20, 2026	Eugen Starschich	Full comparison of the lumped and exact two-port parameters against length; cascaded sections and the equivalent pi; interconnection of two-ports in Appendix A

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1 Abbreviations

Abbreviation	Meaning
AAAC	All-aluminium alloy conductor
AAC	All-aluminium conductor
AC	Alternating current
ACSR	Aluminium conductor steel reinforced
ACSS	Aluminium conductor steel supported
ASTM	American Society for Testing and Materials
AVR	Automatic voltage regulator
CB	Circuit breaker
CIGRE	International Council on Large Electric Systems
DC	Direct current
DS	Disconnector (disconnect switch)
ESDD	Equivalent salt deposit density
GMD	Geometric mean distance
GMR	Geometric mean radius
HVDC	High-voltage direct current
IEC	International Electrotechnical Commission
MOV	Metal-oxide varistor
NSDD	Non-soluble deposit density
OHL	Overhead line
PSS	Power system stabiliser
SCL	Short-circuit level
SCR	Short-circuit ratio
SIL	Surge-impedance loading
SPS	Site pollution severity
SSCI	Sub-synchronous control interaction
SSR	Sub-synchronous resonance
TB	Technical brochure
TS	Technical specification
USCD	Unified specific creepage distance
VSC	Voltage-sourced converter
XLPE	Cross-linked polyethylene

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2 Symbols

Quantities per unit length carry a prime, R', L', G', C'; the same letter without the prime is the quantity of the whole line. Complex quantities are underlined. Terminal quantities take the subscript S at the sending end and R at the receiving end, and voltages are line-to-line unless stated otherwise.

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Symbol	Description
$\underline{A}, \underline{B}, \underline{C}, \underline{D}$	Two-port (ABCD) parameters of the line
C'	Capacitance per unit length
d	Conductor diameter
D	Phase spacing of the tower head
f	System frequency; f_{er} the electrical resonance frequency
G'	Conductance per unit length
GMD	Geometric mean distance between the phases, D_m
GMR	Geometric mean radius of a sub-conductor; GMR_b of the bundle
H	Inertia constant of a synchronous machine
I	Phase current; $\underline{I}_S$, $\underline{I}_R$ at the terminals
k	Degree of series compensation, $X_C = k \cdot X_L$
K_s	Synchronising power coefficient, $\partial P / \partial \delta$
l	Line length; x the distance from the sending end
L'	Inductance per unit length
m	Degree of shunt compensation
n	Turns ratio of an ideal transformer; number of sections in a cascade
P	Active power; P_{nat} natural power, P_{max} maximum transfer
Q	Reactive power
r_b	Equivalent radius of the bundle for the capacitance
R'	Resistance per unit length
s	Bundle spacing
S_{sc}	Short-circuit level at a terminal
U	Line-to-line voltage; $\underline{U}_S$, $\underline{U}_R$ at the terminals, U_n nominal
X	Series reactance of the path; X' per unit length, X_C of the series bank, X_{tot} of the whole path between the internal voltages
$\underline{Y}'$	Shunt admittance per unit length, $G' + j \cdot \omega \cdot C'$
$\underline{Z}'$	Series impedance per unit length, $R' + j \cdot \omega \cdot L'$
$\underline{T}$	Chain (ABCD) matrix of a two-port
$\underline{Z}_{eq}$	Series branch of the equivalent π; $\underline{Y}_{eq}$ its shunt branch
$\underline{Z}_c$	Image (characteristic) impedance of a two-port, $\sqrt{\underline{B}/\underline{C}}$
$\underline{Z}_{in}$	Impedance seen at port 1 of a terminated two-port
$\underline{Z}_L$	Impedance terminating a two-port
$\underline{Z}_w$	Wave (surge) impedance, $\sqrt{\underline{Z}'/\underline{Y}'}$
α	Attenuation constant
β	Phase constant; $\beta \cdot l$ the electrical length
$\underline{\gamma}$	Propagation constant, $\alpha + j \cdot \beta$
δ	Transmission angle between the terminal voltages
ε_0	Permittivity of free space
μ_0	Permeability of free space
$\underline{\theta}$	Image transfer constant of a two-port; $\underline{\gamma} \cdot l$ for a line
φ	Power-factor angle
ω	Angular frequency, $2 \cdot \pi \cdot f$

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3 Introduction

This document collects design notes for VSC HVDC transmission. Each topic is worked out here first, with its derivations, assumptions and sources; the corresponding slides of the HVDC design presentation are then derived from it.

Section **??** covers the creepage distance of insulators for DC overhead lines: why pollution, not overvoltage, sets it, how the site pollution is measured, and how the measured pollution, the insulator type and the altitude lead to the creepage of a 400 kV DC line.

The numerical values given are indicative ranges from the standards and published guidance. They are not agreed AmWatt design constants; a project value follows from the measured site pollution and the supplier's test data.

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4 AC Power Transmission

The familiar statement that active power flows on the angle between two voltages, $P = U_S \cdot U_R / X \cdot \sin\delta$, is a special case. It holds for a short, lossless line, and the error it carries grows with length until it is useless. This section builds the representation from the line parameters upwards: the four per-unit-length quantities, the two-port that follows from them, the exact power transfer relation, the conditions under which it collapses to the simple form, and the distributed solution that replaces the lumped one on long lines.

Nothing in this section distinguishes an overhead line from a cable. The two differ only in the values of the four parameters, and in what those values make critical: Section 4.2 covers the overhead line, Section 4.3 the cable, and Section 4.4 carries the whole chain through on one worked example.

4.1 Fundamentals

The representation below is general: it applies to an overhead line and to a cable alike, and the rest of this chapter specialises it.

4.1.1 Line parameters

A transmission line is described by four parameters per unit length, two in series with the conductor and two in shunt to the return path. Each corresponds to one physical effect, Table 1.

	Parameter	Physical effect
R'	Resistance	Ohmic loss in the conductors
L'	Inductance	Magnetic field around the conductors
G'	Conductance	Leakage through the insulation and corona
C'	Capacitance	Electric field to the other conductors and to earth

Table 1: Line parameters per unit length

They enter as a series impedance and a shunt admittance per unit length,

$$\underline{Z}' = R' + j \cdot \omega \cdot L' \qquad \underline{Y}' = G' + j \cdot \omega \cdot C' \tag{1}$$

R' and L' follow from the conductor itself and from the arrangement of the conductors; G' and C' from the insulation between them. The expressions differ between an overhead line and a cable and are given in Sections 4.2 and 4.3.

Taken over a finite length the four parameters arrange themselves as the nominal π-section of Figure 1: the series branch carries R' and L', and the shunt branch G' and C' is split equally between the two ends.

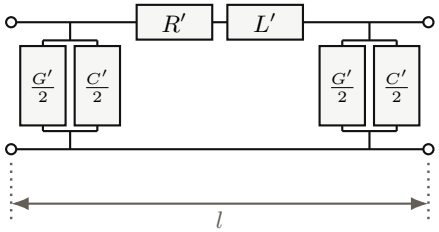


Figure 1: Nominal π-section of a line of length l, written in the per-unit-length parameters

4.1.2 The line as a two-port

Over a length l the parameters give a series impedance and a shunt admittance

$$\underline{Z} = \underline{Z}' \cdot l \qquad \underline{Y} = \underline{Y}' \cdot l \tag{2}$$

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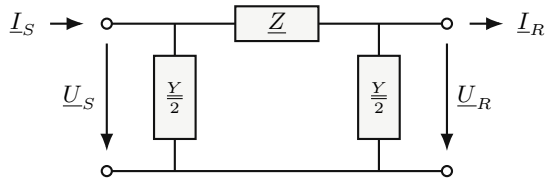


Figure 2: The same π-section as a two-port: the whole series impedance $\underline{Z}$ between the rails, half the shunt admittance $\underline{Y}$ at each end

which the nominal π arranges as in Figure 2: the whole series impedance between the rails and half the shunt admittance at each end. This is the standard two-port description of a linear network, of which the line is one case; Appendix A gives the general form for the reader who wants it, and is not needed here. The ports are named by function throughout this section, the sending end and the receiving end. The line is then a two-port,

$$\begin{bmatrix} \underline{U}_S \\ \underline{I}_S \end{bmatrix} = \begin{bmatrix} \underline{A} & \underline{B} \\ \underline{C} & \underline{D} \end{bmatrix} \cdot \begin{bmatrix} \underline{U}_R \\ \underline{I}_R \end{bmatrix} \tag{3}$$

with, for the nominal π,

$$\underline{A} = \underline{D} = 1 + \frac{\underline{Z} \cdot \underline{Y}}{2}, \qquad \underline{B} = \underline{Z}, \qquad \underline{C} = \underline{Y} \cdot \left(1 + \frac{\underline{Z} \cdot \underline{Y}}{4}\right) \tag{4}$$

The shunt admittance shows itself in $\underline{A}$. A line that did not charge itself would have $\underline{A} = 1$ and would behave as a single series impedance; how far $|\underline{A}|$ departs from unity is a measure of how far it is from that.

4.1.3 Power transfer

Solving (3) for the receiving-end current gives

$$\underline{I}_R = \frac{\underline{U}_S - \underline{A} \cdot \underline{U}_R}{\underline{B}} \tag{5}$$

Taking the receiving-end voltage as the phase reference, so that δ is the angle by which the sending-end voltage leads it,

$$\underline{U}_S = U_S \cdot e^{j \cdot \delta}, \quad \underline{U}_R = U_R, \quad \underline{A} = |A| \cdot e^{j \cdot \vartheta_A}, \quad \underline{B} = |B| \cdot e^{j \cdot \vartheta_B} \tag{6}$$

the receiving-end complex power $\underline{S}_R = \underline{U}_R \cdot \underline{I}_R^*$ separates into

$$P_R = \frac{U_S \cdot U_R}{|B|} \cdot \cos\left(\vartheta_B - \delta\right) - \frac{|A| \cdot U_R^2}{|B|} \cdot \cos\left(\vartheta_B - \vartheta_A\right) \tag{7}$$

$$Q_R = \frac{U_S \cdot U_R}{|B|} \cdot \sin\left(\vartheta_B - \delta\right) - \frac{|A| \cdot U_R^2}{|B|} \cdot \sin\left(\vartheta_B - \vartheta_A\right) \tag{8}$$

with U_S and U_R line-to-line magnitudes, so that P_R and Q_R are three-phase quantities.

Two numbers carry the whole behaviour of the line. The angle ϑ_B sets where the maximum of P_R falls: differentiating (7) puts it at $\delta = \vartheta_B$, which is $90°$ only for a lossless line. The magnitude $|A|$ scales the second term, a constant offset independent of δ that exists only because the line charges itself.

4.1.4 The short, lossless line

For a short line the shunt admittance vanishes and $\underline{A} = \underline{D} = 1$, so $|A| = 1$ and $\vartheta_A = 0$. For a lossless line $\underline{B} = \underline{Z} = j \cdot X$, so $|B| = X$ and $\vartheta_B = 90°$. With $\cos\left(90° - \delta\right) = \sin\delta$ and $\cos 90° = 0$, equations (7) and (8) collapse to

$$P = \frac{U_S \cdot U_R}{X} \cdot \sin\delta \qquad Q_R = \frac{U_S \cdot U_R \cdot \cos\delta - U_R^2}{X} \tag{9}$$

with the series reactance

$$X = \omega \cdot L' \cdot l \tag{10}$$

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For small angles this separates further into $P \approx U_S \cdot U_R \cdot \delta / X$ and $Q_R \approx U_R \cdot (U_S - U_R) / X$: active power is carried by the angle between the terminal voltages, reactive power by the difference in their magnitude. That decoupling is what makes (9) worth keeping as a mental model even where it is not accurate enough to compute with.

The X of (9) is not the line alone but the whole path between the two internal voltages — the source impedances and the transformers at both ends are in series with it — so the short-circuit level at the terminals belongs in any statement of transfer capability.

4.1.5 Distributed representation

The π-section lumps parameters that are in fact distributed along the line. The distributed description uses two quantities,

$$\underline{\gamma} = \alpha + j \cdot \beta = \sqrt{\underline{Z}' \cdot \underline{Y}'} \qquad \underline{Z}_w = \sqrt{\frac{\underline{Z}'}{\underline{Y}'}} \tag{11}$$

the propagation constant and the wave impedance. Voltage and current are then defined at every cross-section of the line rather than only at its terminals, Figure 3. At a distance x from the sending end of a line of length l they are

$$\underline{U} = \underline{U}_R \cdot \cosh\left(\underline{\gamma} \cdot (l - x)\right) + \underline{Z}_w \cdot \underline{I}_R \cdot \sinh\left(\underline{\gamma} \cdot (l - x)\right) \tag{12}$$

$$\underline{I} = \frac{\underline{U}_R}{\underline{Z}_w} \cdot \sinh\left(\underline{\gamma} \cdot (l - x)\right) + \underline{I}_R \cdot \cosh\left(\underline{\gamma} \cdot (l - x)\right) \tag{13}$$

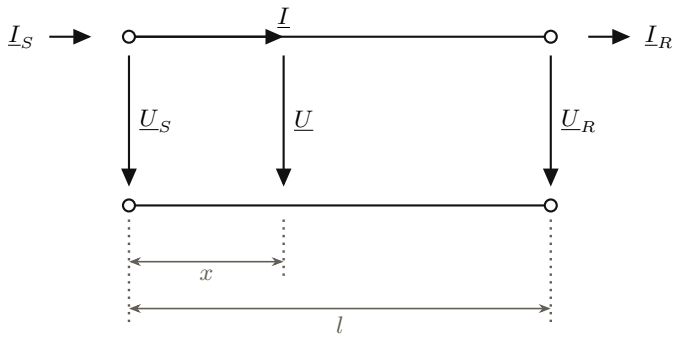


Figure 3: Terminal quantities of the two-port and the local voltage and current at a distance x along the line

Evaluating (12) and (13) at $x = 0$ and comparing with (3) identifies the exact two-port parameters,

$$\underline{A} = \underline{D} = \cosh\left(\underline{\gamma} \cdot l\right), \qquad \underline{B} = \underline{Z}_w \cdot \sinh\left(\underline{\gamma} \cdot l\right), \qquad \underline{C} = \frac{\sinh\left(\underline{\gamma} \cdot l\right)}{\underline{Z}_w} \tag{14}$$

which replace the nominal-π entries of (4). The relations (7) and (8) are unchanged; only the way $\underline{A}$ and $\underline{B}$ are computed differs.

The exact result can itself be written as a π. Matching the branches of a π-section to (14) entry by entry gives

$$\underline{Z}_{eq} = \underline{Z}_w \cdot \sinh\left(\underline{\gamma} \cdot l\right), \qquad \frac{\underline{Y}_{eq}}{2} = \frac{1}{\underline{Z}_w} \cdot \tanh\left(\frac{\underline{\gamma} \cdot l}{2}\right) \tag{15}$$

the *equivalent* π, often called the long-line correction. It is not an approximation: one π with these two branches reproduces the distributed line exactly at its terminals at that frequency, because $1 + \sinh\left(\underline{\gamma} \cdot l\right) \cdot \tanh\left(\underline{\gamma} \cdot l/2\right) = \cosh\left(\underline{\gamma} \cdot l\right)$. The nominal π of (4) is its small-argument limit, with $\sinh u \to u$ and $\tanh u \to u$. This is what a load-flow program stores for a long line, and it is why a line keeps a π-shaped model however long it is. Section 4.4.3 puts numbers on both.

What decides whether the lumped form will do is the electrical length $\beta \cdot l$, not the length in km. Since $\beta = \omega \cdot \sqrt{L' \cdot C'}$ for a low-loss line, a cable reaches a given electrical length in a fraction of the distance an overhead line needs, and the lumped approximation fails correspondingly earlier.

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4.2 Overhead Lines

4.2.1 Geometry and the line parameters

On an overhead line the conductors are bare and the insulation is air, so the inductance and the capacitance follow from the geometry of the tower and of the conductor bundle alone:

$$L' = \frac{\mu_0}{2 \cdot \pi} \cdot \ln \frac{D_m}{r_b} \qquad C' = \frac{2 \cdot \pi \cdot \varepsilon_0}{\ln \left(D_m / r_{b_c} \right)} \tag{16}$$

with D_m the geometric mean distance between the phases and r_b, r_{b_c} the equivalent radii of the bundle for the inductance and the capacitance respectively. The two differ because the inductance is referred to the geometric mean radius of the sub-conductors and the capacitance to their physical radius.

Both parameters depend on the geometry only through a logarithm, and on the same ratio D_m / r_b. That makes the geometry a weak lever: the phase spacing and the bundle radius have to be changed by a factor to move L' or C' by a fraction. Length, by contrast, enters linearly, so on an overhead line the series reactance of (10) is set by the route far more than by the tower.

The flat (horizontal) configuration used throughout this report is drawn to scale in Figure 4: three phases on one crossarm at a spacing D, so that the outer phases see $2 \cdot D$, each phase a bundle of three sub-conductors. Table 2 gives its dimensions. It gives the lowest towers and the widest right-of-way of the common arrangements.

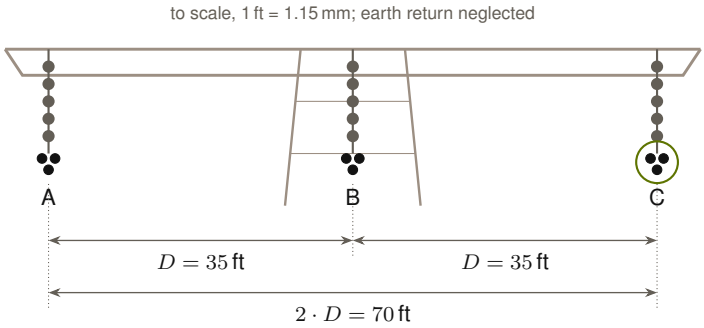


Figure 4: Tower head of the reference line, flat configuration; the marked phase is drawn in detail in Figure 5. Tower body and insulator lengths are schematic

Parameter		Value
D phase spacing	ft	35
$2 \cdot D$ outer phases	ft	70
Sub-conductors per phase		3
s bundle spacing	in	18
d conductor diameter	in	1.762
GMR of a sub-conductor	ft	0.0588

Table 2: Tower-head geometry of the reference line, Bluebird ACSR 2156 kcmil 84/19, single circuit [10]

Bundling is the one geometric change that is worth making for its electrical effect. Dividing the phase current between sub-conductors raises the equivalent radius r_b well above that of a single conductor, and with it reduces L'; it also reduces the surface gradient and so the corona loss. Table 3 shows the effect for one configuration: going from one sub-conductor to three removes a third of the reactance, while doubling r_b again — or halving D_m — removes only a further 15 %.

The shunt branch of an overhead line is effectively a pure capacitance. The conductance G' collects the leakage across the insulator strings and the corona loss, and in normal weather it is two orders of magnitude below the susceptance $\omega \cdot C'$, so it is set to zero in load-flow work. It is not negligible for loss studies in foul weather, where corona dominates it.

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Sub-conductors per phase	r_b	X'	relative
	m	Ω/km	%
1	0.018	0.499	148
2	0.091	0.377	112
3	0.155	0.336	100
4	0.222	0.310	92

Table 3: Effect of bundling on the series reactance, 35 ft flat configuration with 18 in bundle spacing at 60 Hz

Because the conductors are far apart and surrounded by air, C' stays small. The wave impedance $\underline{Z}_w = \sqrt{\underline{Z}'/\underline{Y}'}$ is correspondingly high — a few hundred ohms — and the charging current per unit length is low enough that an overhead line can run for hundreds of kilometres before it has to be compensated.

4.2.2 Bundle and phase equivalents

The two radii of (16) and the mean phase distance follow from the tower head and the bundle. For a three-conductor bundle of spacing s made of sub-conductors of diameter d and geometric mean radius GMR, Figure 5, and a flat configuration of phase spacing D in which the outer phases see $2 \cdot D$,

$$GMR_b = \sqrt[3]{GMR \cdot s^2}, \qquad r_b = \sqrt[3]{\frac{d}{2} \cdot s^2}, \qquad GMD = \sqrt[3]{D \cdot D \cdot 2 \cdot D} = D \cdot \sqrt[3]{2} \tag{17}$$

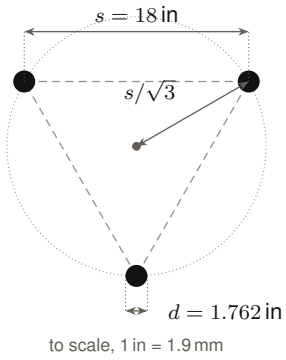


Figure 5: Three-conductor bundle of the reference line, apex down, with the circle of radius $s/\sqrt{3}$ on which the sub-conductors sit

with GMR_b the radius entering $\underline{L}'$ and r_b the one entering C'. For a two-conductor bundle the same construction gives $GMR_b = \sqrt{GMR \cdot s}$ and $r_b = \sqrt{d/2 \cdot s}$. The GMD above assumes a transposed line.

Table 4 evaluates them for the reference tower head of Section 4.4 — 35 ft flat, 18 in bundle spacing, Bluebird ACSR 2156 kcmil with $d = 1.762$ in and $GMR = 0.0588$ ft — with three and with two sub-conductors per phase.

The third sub-conductor lowers Z_w by about 10 % and so raises the natural power of Section 4.5 by about 11 %. A flat configuration gives the lowest towers and the widest right-of-way; the outer phases see twice the phase spacing, which is what the cube root in (17) averages.

4.2.3 Conductor codewords

A conductor size does not define a conductor. ACSR of 795 kcmil exists in four standard strandings — Cuckoo 24/7, Drake 26/7, Mallard 30/19 and Condor 54/7 — each with a different steel content and therefore different strength, weight and sag. The codeword fixes area, stranding and core in one word.

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Quantity		$3\times$	$2\times$
GMR_b	in	6.11	3.56
r_b	in	6.58	3.98
GMD	ft	44.1	44.1
L'	mH/km	0.892	1.000
C'	nF/km	12.68	11.38
$Z_w = \sqrt{L'/C'}$	Ω	265	296

Table 4: Bundle equivalents and the resulting parameters, three against two sub-conductors per phase

The names began as Alcoa product nomenclature in the 1920s and 1930s and were later adopted into the ASTM standards [13]. The family identifies the construction: ACSR and ACSS take bird names (Drake, Falcon, Rail, Cardinal), AAC flowers (Rose, Iris, Tulip, Peony) [14] and AAAC US city names (Akron, Amherst, Anaheim) [15]. The name carries no size information — Bluebird is 2156 kcmil, Condor only 795.

Other systems use the same idea with different animals. BS 215 and Indian practice use mammals: Dog, Wolf, Lynx, Panther, Zebra, Moose [16]. A 400 kV Indian line specified as “twin Moose” is the equivalent of a North American “bundled Drake”. Metric designs in continental Europe usually drop codewords altogether and quote the aluminium and steel areas directly, for example 490/65 mm^2.

The practical consequence is that a conductor has to be quoted with its standard as well as its codeword, since the same name can appear in more than one national system.

4.3 Cables

A cable carries the same four parameters and the same representation, but the geometry inverts the shunt branch. Conductor and screen are separated by a few centimetres of solid dielectric rather than by metres of air, so the capacitance per unit length rises by more than an order of magnitude while the inductance falls, Table 5.

Parameter		OHL	Cable
R'	Ω/km	0.1–1	0.1–1
L'	mH/km	≈ 1	0.31–1.42
G'	µS/km	0.05	≈ 1
C'	nF/km	≈ 10	180–420

Table 5: Indicative per-unit-length parameters, overhead line against XLPE cable [17]

Three consequences follow, and they are what make a cable a different design problem rather than a shorter overhead line.

The wave impedance collapses. With C' up and L' down, $\underline{Z}_w$ falls from a few hundred ohms to under a hundred. For a 230 kV XLPE cable with $L' = 1.35$ mH/km and $C' = 190$ nF/km, $\underline{Z}_w = 84\,\Omega$ against $266\,\Omega$ for the overhead line of Section 4.4. The natural power rises in proportion, but it is not a loading anyone reaches: the limit arrives from the charging current first.

The line charges itself heavily. The charging reactive power per unit length is $Q'_c = U^2 \cdot \omega \cdot C'$. For the same example cable $\omega \cdot C' = 71.6$ µS/km, giving 3.79 MVAr/km and a charging current of 9.5 A/km at 230 kV. That current flows whether or not the cable is loaded, it is drawn from both ends, and it consumes conductor ampacity that is then unavailable for useful load. Beyond a critical length the cable carries nothing but its own charging current. Shunt reactors at the ends — and, for long routes, at intermediate points — are therefore not an optimisation but a condition of operation.

The electrical length runs out sooner. The phase constant $\beta = \omega \cdot \sqrt{L' \cdot C'}$ rises with the capacitance, so a cable reaches a given $\beta \cdot l$ in a fraction of the route length an overhead line needs. The example cable has a wavelength of 1041 km against 4936 km for the reference overhead line: one kilometre of this cable is electrically worth nearly five of that line. The lumped π of Section 4.1.2 therefore loses accuracy at a few tens of kilometres rather than a few hundred, and the distributed form of (14) has to be used much earlier.

Where intermediate compensation is impossible — a submarine crossing, or a long cable with no accessible landing point — these three together are what rule out AC and leave DC as the only option. The comparison is then not between a cable and an overhead line but between an AC and a DC scheme.

4.4 Worked Example: the 500 kV Reference Line

Most of the numbers in this report come from one line. It is a 500 kV, 60 Hz overhead line of 300 mi (483 km) in flat configuration, with three Bluebird ACSR 2156 kcmil sub-conductors per phase, 35 ft phase spacing and 18 in bundle spacing. The conductor data are the Southwire ACSR tables [10] and the resistance is the value at 75 °C. Table 6 gives the line and the parameters that follow from it by (16). Section 4.5 uses a second line — 400 kV, 50 Hz, 400 km — and says so where it does.

Quantity		Value
U_n	kV	500
f	Hz	60
l	mi (km)	300 (483)
Conductor		$3\times$ Bluebird ACSR
Geometry	ft / in	35 flat / 18 bundle
R' (75 °C)	mΩ/km	56.9
L'	mH/km	0.891
$X' = \omega \cdot L'$	Ω/km	0.336
G'	µS/km	0.05
C'	nF/km	12.7

Table 6: 500 kV reference line and its per-unit-length parameters

At 60 Hz the susceptance is $\omega \cdot C' = 4.79$ µS/km, about a hundred times the conductance, which is the overhead-line case of Section 4.2: the shunt branch is treated as a pure capacitance throughout.

4.4.1 Line constants

Table 7 collects the lumped and distributed constants of the whole 300 mi line.

4.4.2 What the lumped approximation costs

Table 8 evaluates all four two-port parameters of this line at several lengths, once as the nominal π of (4) and once exactly from (14); $\underline{D} = \underline{A}$ in both. Table 9 gives the deviations, and Figure 6(a) the same comparison continuously against length [12].

The four parameters do not degrade together, and the differences matter for different reasons.

- $|\underline{B}|$ is the worst and errs high, because the nominal π keeps the whole series impedance in one branch while the real line distributes it among the shunt capacitance. Since P_{max} goes inversely with $|\underline{B}|$, the lumped model is *pessimistic* about transfer capability.
- $|\underline{A}|$ and $|\underline{C}|$ err low, and $|\underline{A}|$ stays within a percent to about 350 mi — so a model judged on its voltage ratio alone looks far better than it is.

Quantity		Value
$\underline{Z}$	Ω	$27.5 + j\,162.2$
$\underline{Y}$	mS	$j\,2.31$
Charging at 1.0 p.u.	MVAr	578
α	Np/km	1.07×10^{-4}
β	rad/km	1.27×10^{-3}
$\beta \cdot l$	°	35.2
Wavelength	km	4936
$\underline{Z}_w$	Ω	$266 - j\,22$
$P_{nat} = U_n^2/\mathrm{Re}\{\underline{Z}_w\}$	MW	939

Table 7: Computed quantities of the reference line at 300 mi

- P_{max} collects both errors and lands between them: -7.2 % at 300 mi, -26 % at 600.

At 300 mi the nominal π is still usable for voltage work and already misleading for capability. Beyond about 400 mi nothing about it is defensible.

l	$\beta \cdot l$	$\lvert\underline{A}\rvert$		$\lvert\underline{B}\rvert$		$\lvert\underline{C}\rvert$	
mi	°	π	exact	π, Ω	exact, Ω	π, mS	exact, mS
100	11.7	0.979	0.979	54.8	54.5	0.762	0.765
200	23.5	0.917	0.918	109.7	106.7	1.477	1.499
300	35.2	0.813	0.819	164.5	154.4	2.095	2.170
400	46.9	0.669	0.686	219.4	195.8	2.570	2.751
600	70.4	0.280	0.351	329.1	252.9	2.904	3.553

Table 8: Two-port parameters of the reference line against length, nominal π and exact; $\underline{D} = \underline{A}$ throughout [12]

l	$\beta \cdot l$	$\lvert\underline{A}\rvert$	$\lvert\underline{B}\rvert$	$\lvert\underline{C}\rvert$	P_{max}
mi	°	%	%	%	%
100	11.7	-0.0	$+0.7$	-0.4	-0.8
200	23.5	-0.1	$+2.8$	-1.4	-3.3
300	35.2	-0.7	$+6.5$	-3.4	-7.2
400	46.9	-2.5	$+12.0$	-6.6	-12.4
600	70.4	-20.0	$+30.1$	-18.2	-25.8

Table 9: Deviation of the nominal π from the exact two-port, and the resulting error in the maximum transfer at 500 kV [12]

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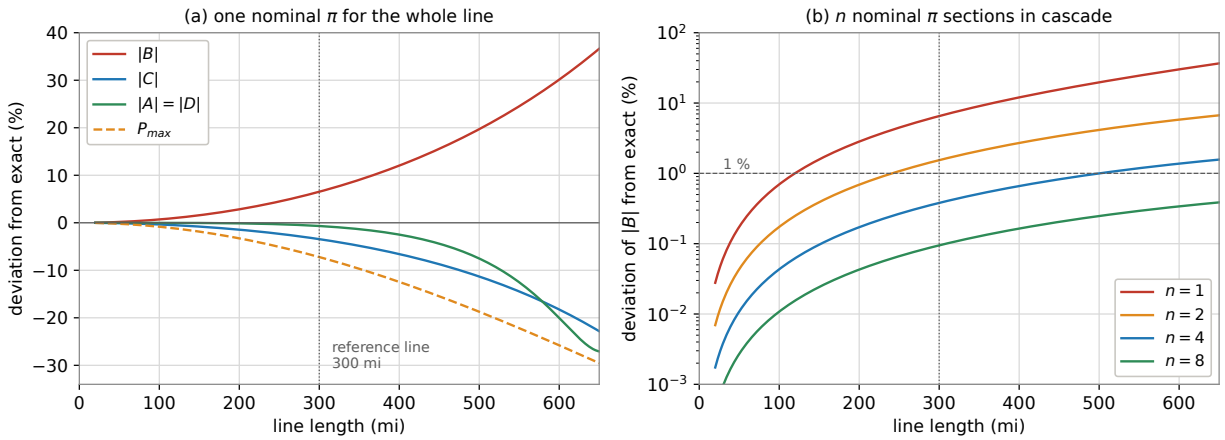


Figure 6: (a) Deviation of the nominal π against line length; (b) the same deviation of $|\underline{B}|$ when the line is built from n cascaded π-sections. Both stop short of the quarter wavelength, where $|\underline{A}|$ passes through zero

4.4.3 Cascading shorter sections

The nominal π fails because one lumped section cannot represent a distribution. The obvious repair is to use more of them: cut the line into n pieces of length l/n, model each as its own nominal π, and connect the two-ports in cascade. The chain matrices multiply, so the whole line is

$$\underline{T} = \left[\underline{T}_{\pi}\left(\frac{l}{n}\right)\right]^{n} \tag{18}$$

which is one matrix power, not n separate models. Each section is shorter, so each is a better approximation, and in the limit $n \to \infty$ the product is the distributed solution of (14) — the cascade is how the exact form is reached from the lumped one.

Table 10 and Figure 6(b) show the convergence. The deviation falls with $1/n^2$ once the sections are electrically short, so each doubling of n buys a factor of four. What governs the result is the electrical length of a *section*, not of the line: on this conductor, sections of about 100 mi hold $|\underline{B}|$ inside a percent at any total length, which is the same 100 mi that Table 8 gives for a whole line.

n	**section**	deviation of $\lvert\underline{B}\rvert$, %				
		100 mi	200 mi	300 mi	400 mi	600 mi
1	l	$+0.70$	$+2.83$	$+6.53$	$+12.03$	$+30.13$
2	$l/2$	$+0.17$	$+0.69$	$+1.54$	$+2.71$	$+5.81$
4	$l/4$	$+0.04$	$+0.17$	$+0.38$	$+0.66$	$+1.38$
8	$l/8$	$+0.01$	$+0.04$	$+0.09$	$+0.16$	$+0.34$
16	$l/16$	$+0.00$	$+0.01$	$+0.02$	$+0.04$	$+0.08$

Table 10: Nominal π-sections in cascade: deviation of $|\underline{B}|$ from the exact two-port against the number of sections [12]

Cascading is not the only repair, and for a plain line it is not the best one. The equivalent π of (15) is exact with a single section, and the correction it applies is not small: on this line at 300 mi the series branch falls from $27.5 + j\,162.2\,\Omega$ to $24.1 + j\,152.5\,\Omega$ and the shunt branch rises from 1.156 to 1.193 mS at each end, and at 600 mi from $54.9 + j\,324.4$ to $30.4 + j\,251.0\,\Omega$ against 2.312 to 2.650 mS. One exact section beats sixteen approximate ones and costs less to evaluate.

Cascading earns its place where the single equivalent π cannot go.

- **Something is connected part-way along.** A midpoint series bank is a two-port cascaded between two half-lines, Section 4.8, and a tapped substation or a transposition point divides the line in the same way. The equivalent π describes a line between its own two ends and nothing else.

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- **Quantities along the line are wanted.** The voltage profiles of Section 4.5.3 are the intermediate port quantities of a cascade; a single two-port gives only its terminals.
- **The line is not uniform.** Different conductor, spacing or bundle over part of the route gives each part its own matrix, and the product handles it with no extra theory. The equivalent π assumes one $\underline{\gamma}$ and one $\underline{Z}_w$ over the whole length.
- **A time-domain or frequency-sweep model is wanted.** The cascade is a lumped circuit a simulator can integrate directly, which is why electromagnetic-transient models still use chained sections rather than a closed-form $\cosh$.

Where the line is uniform and only its terminals matter, the equivalent π is the right model, and the cascade is a way of understanding where it comes from. Appendix A.4 sets the cascade beside the other ways two-ports combine, which are not interchangeable with it.

4.4.4 The three representations compared

Table 11 evaluates the two-port parameters and the crest of the power-angle curve for the same line under the three representations of Section 4.1. The simplification of (9) moves the maximum from $81.0°$ to $90°$ and overstates it by 13 %; the nominal π keeps the angle but understates the maximum by 7 %, because its $|\underline{B}|$ is too large. At 300 mi $|\underline{A}| = 0.82$, so the line is already far from the single series impedance that (9) assumes. The numbers follow from the model of Section 4.1 implemented in [11].

Model	$\lvert A\rvert$	ϑ_B	$\lvert B\rvert$	P_{max}
		°	Ω	MW
Short, lossless	1.000	90.0	162.2	1541
Nominal π	0.813	80.4	164.5	1266
Distributed	0.819	81.0	154.4	1364

Table 11: Two-port parameters and maximum transfer at 300 mi, both terminals at 1.0 p.u.

4.4.5 The surrounding system

The reactance that appears in (9) is the whole path between the two internal voltages, not the line:

$$X_{tot} = X_{th_s} + X_{tr_s} + X' + X_{tr_r} + X_{th_r}, \qquad X_{th} = \frac{U_n^2}{S_{sc}}, \qquad \mathrm{SCR} = \frac{S_{sc}}{P_n} \tag{19}$$

with X_{th} the Thevenin reactance of the system behind a terminal, X_{tr} that of the transformer and S_{sc} the short-circuit level there. Taking the lossless $P_{max} = U_n^2/X_{tot}$ with equal systems at both ends and the transformers neglected, Table 12 shows how much of the capability belongs to the terminals rather than to the line.

SCL, each end	X_{th}	X_{tot}	P_{max}
GVA	Ω	Ω	MW
5	50.0	253.4	986
10	25.0	203.4	1229
20	12.5	178.4	1401
stiff	0.0	153.4	1630

Table 12: Effect of the surrounding system on the maximum transfer of the 300 mi reference line, $X' = 153.4\,\Omega$, transformers neglected

A quarter of the capability of this line into 10 GVA systems is lost to the terminals rather than to the line itself, which is why a loadability figure quoted for a line in isolation is optimistic. The same quantity, normalised as the short-circuit ratio, is what governs converter performance on a DC link.

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4.4.6 What this is not

P_{max} is not a transfer capability. It is the crest of the power-angle curve of an uncompensated line between two stiff systems, reached at an angle no system is operated at. What the line can actually carry follows from the thermal rating of the conductors, the permissible voltage drop and the angular stability margin, and from the contingency case rather than the intact one. Those limits are the subject of a separate section; the representation of Section 4.1 is what they are evaluated on.

4.5 Natural Power

4.5.1 Definition

A line terminated in its own wave impedance carries its *natural power*: the reactive power generated by the shunt capacitance exactly cancels the reactive power consumed by the series inductance, the line exchanges no reactive power with either end, and its voltage magnitude stays constant along the route. North American practice calls the same quantity the surge-impedance loading, SIL.

On an overhead line $\omega \cdot L' \gg R'$ and $\omega \cdot C' \gg G'$, so the wave impedance of (11) is within about a percent of the real quantity $\sqrt{L'/C'}$, and it is that real value the natural power is quoted from:

$$Z_w \approx \sqrt{\frac{L'}{C'}} \qquad P_{nat} = \mathrm{SIL} = \frac{U^2}{Z_w} \tag{20}$$

Z_w is a property of the conductor geometry alone. It does not depend on the length of the line, and in the lossless form not on frequency either. P_{nat} is therefore a reference rather than a rating: below it the line is a net source of reactive power, above it a net sink.

The same result follows from the two-port equations. Closing the line with $\underline{Z}_w$ makes $\underline{I}_R = \underline{U}_R / \underline{Z}_w$, and (12) and (13) collapse to a single travelling wave with no reflection,

$$\underline{U}(x) = \underline{U}_R \cdot e^{\underline{\gamma} \cdot x}, \qquad \underline{I}(x) = \underline{I}_R \cdot e^{\underline{\gamma} \cdot x} \tag{21}$$

so that $\underline{U}/\underline{I} = \underline{Z}_w$ everywhere and the magnitude falls only by the attenuation $e^{-\alpha \cdot l}$, Figure 7. Only the complex $\underline{Z}_w$ matches perfectly; terminating in the real $\sqrt{L'/C'}$ leaves a small residual reflection.

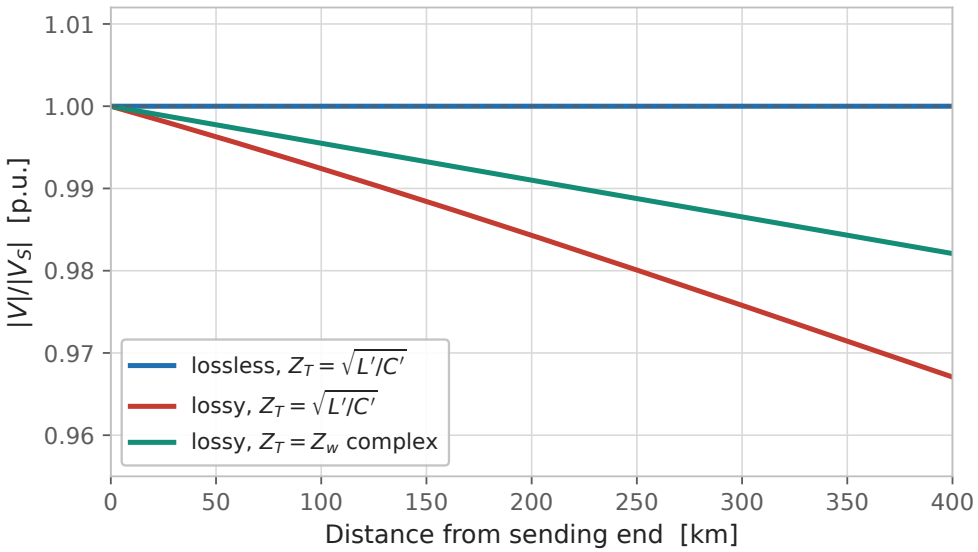


Figure 7: Voltage along a line closed in its wave impedance: the profile is flat apart from the attenuation, and the residual ripple of the real-valued termination

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4.5.2 The reactive balance

Read as a reactive power balance, a line of length l at line-to-line voltage U carrying phase current I generates and consumes

$$Q_C = \omega \cdot C' \cdot l \cdot U^2 \qquad Q_L = 3 \cdot \omega \cdot L' \cdot l \cdot I^2 \tag{22}$$

Natural loading is where the two cancel. Setting $Q_C = Q_L$ eliminates both l and ω and gives $I = U/\left(\sqrt{3} \cdot Z_w\right)$, hence $P_{nat} = \sqrt{3} \cdot U \cdot I = U^2/Z_w$ again. The cancellation is independent of length, which is why the natural power is a property of the cross-section and not of the route.

The worked line of this section is a 400 kV, 50 Hz line of 400 km with a two-conductor bundle, $R' = 25\,\text{m}\Omega\text{/km}$, $X' = 0.30\,\Omega\text{/km}$ and $C' = 12.5\,\text{nF/km}$, giving $L' = 0.955\,\text{mH/km}$, $Z_w = 276.4\,\Omega$, $|\underline{Z}_w| = 276.9\,\Omega$, $\beta \cdot l = 24.88°$ and $P_{nat} = 579\,\text{MW}$. It is not the 500 kV line of Section 4.4.

Losses move the balance. On the lossless line the zero crossing of the net reactive flow reproduces U^2/Z_w to six decimals; on the real line it falls to 566 MW, 2.2 % below. The sending-end current carries the load plus the losses, so $3 \cdot I^2 \cdot X' \cdot l$ is larger and the balance is reached at a lower transfer. Figure 8 plots the net reactive flow of both against the transfer.

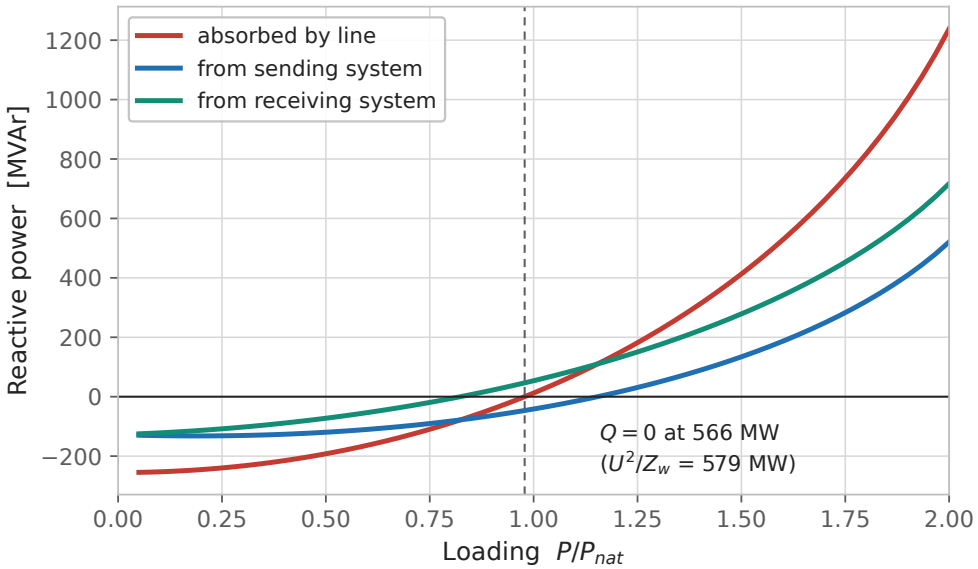


Figure 8: Net reactive power absorbed by the line against transfer, $Q_{line} = \Im\{\underline{S}_S\} - \Im\{\underline{S}_R\}$, lossless and real line

4.5.3 Voltage profile

With both terminals held at 1.0 p.u. and the transmission angle solved for each loading, the profile along the line shows the balance directly, Figure 9. Below P_{nat} the shunt capacitance dominates and the middle of the line rises, to 1.024 p.u. at no load. At P_{nat} the profile is flat to four decimals. Above it the series reactance dominates and the middle sags, to 0.873 p.u. at $2 \cdot P_{nat}$, by which point the angle across the line has reached $63°$.

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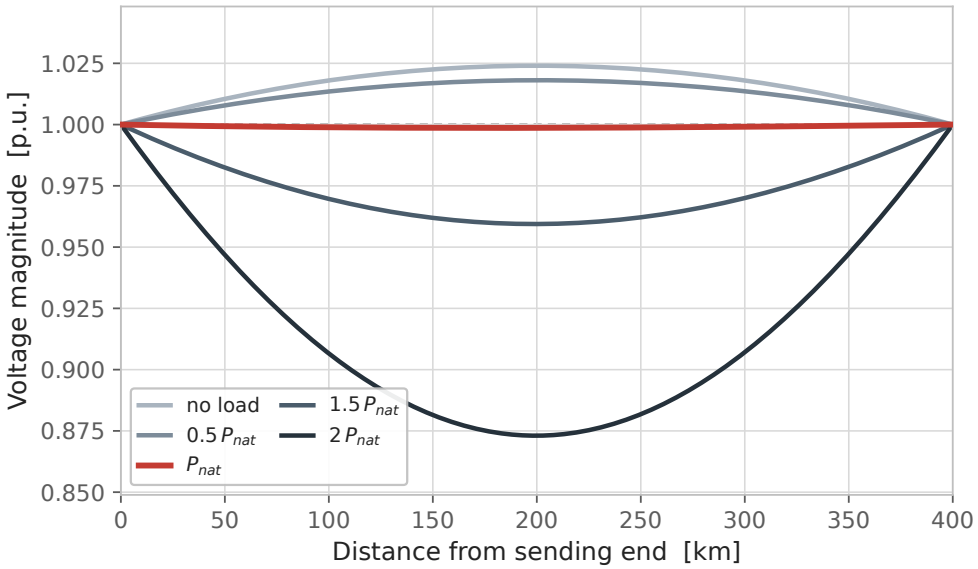


Figure 9: Voltage profile along the 400 km line at several loadings, both terminals at 1.0 p.u.

4.5.4 Typical values

Table 13 gives the natural power of common line types.

Line	U	C'	Z_w	P_{nat}	S_{th}/P_{nat}
	kV	nF/km	Ω	MW	
230 kV, $1\times$ Drake	230	9.0	376.1	141	2.8
345 kV, $2\times$ bundle	345	11.8	292.3	407	2.6
400 kV, $2\times$ bundle	400	12.5	276.4	579	2.4
500 kV, $3\times$ bundle	500	13.5	256.6	974	2.3
765 kV, $4\times$ bundle	765	13.5	250.8	2334	2.0

Table 13: Natural power of typical line types

Three things follow. Z_w spans only 251 to 376 Ω across the whole range while P_{nat} spans 141 to 2334 MW, because P_{nat} is a U^2 quantity: voltage, not geometry, is what buys transfer. Bundling and tighter phase spacing are the only levers on Z_w, and they work on both terms, lowering L' and raising C'. And the ratio of thermal rating to natural power, S_{th}/P_{nat}, decides how far the thermal plateau of the loadability curve of Section 4.6 extends.

AC cables sit at $Z_w \approx 20 \ldots 40\,\Omega$, so their natural power is far above the thermal rating and never reached; as Section 4.3 sets out, the charging current binds first.

4.5.5 Light load: the Ferranti effect

On an open-ended line $\underline{I}_R = 0$ and the terminal voltage ratio follows from (14) as

$$\frac{\underline{U}_R}{\underline{U}_S} = \frac{1}{\cosh\left(\underline{\gamma} \cdot l\right)} \longrightarrow \frac{1}{\cos\left(\beta \cdot l\right)} \tag{23}$$

in the lossless limit, so the receiving end rises above the sending end. At $\beta \cdot l = 24.88°$ the open end reaches 1.102 p.u., drawing 251 MVAr of charging power. A 60 % shunt reactor brings it back to 0.984 p.u.; full compensation gives 0.918 p.u., Figure 10. The degree of compensation follows from the light-load voltage, but it also moves the natural power.

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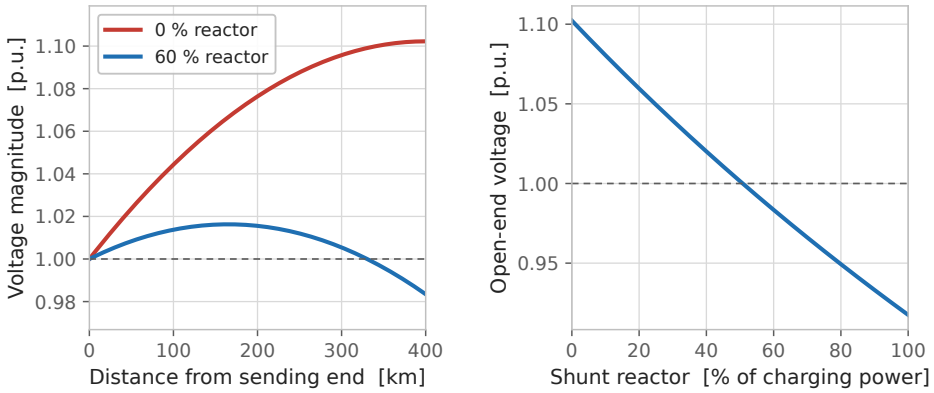


Figure 10: Open-end voltage rise against line length, and the effect of shunt compensation

4.5.6 Compensation moves the natural power

For uniformly distributed compensation the scaling is exact. Series compensation of degree k reduces the reactance and shunt compensation of degree m the capacitance:

$$X' \to X' \cdot (1-k) \;\Rightarrow\; \frac{P_{nat}}{\sqrt{1-k}}, \qquad C' \to C' \cdot (1-m) \;\Rightarrow\; P_{nat} \cdot \sqrt{1-m} \tag{24}$$

so $k = 60\,\%$ raises P_{nat} by a factor 1.581 and $m = 60\,\%$ lowers it by 0.632, Figure 11.

Real banks are lumped at substations rather than distributed, so these laws are a first-order guide only. A lumped 30 % series bank at $1.4 \cdot P_{nat}$ on the worked line reduces the transmission angle from $37.6°$ to $25.6°$ and turns the sending-end reactive flow from $+89$ to -30 MVAr. Section 4.8 treats the lumped case properly.

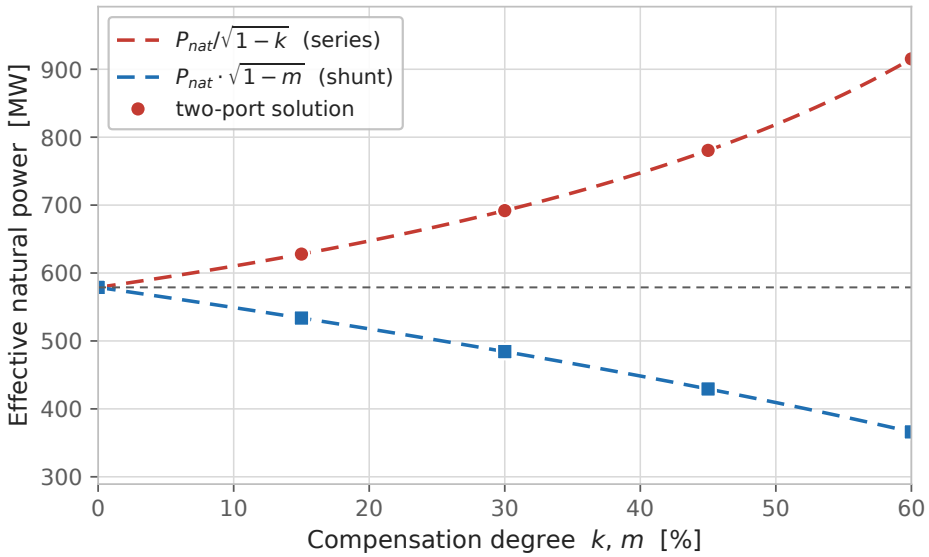


Figure 11: Effect of distributed series and shunt compensation on the natural power

4.6 Transmission Limits

The representation of Section 4.1 says what a line does, not what it may be asked to do. Three criteria bound the transfer, and which of them binds depends almost entirely on length.

4.6.1 The three criteria

Thermal. The conductor temperature sets a current, and with it a transfer that is independent of length:

$$P_{th} = \sqrt{3} \cdot U_n \cdot I_{th} \tag{25}$$

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The limiting mechanism is annealing of the aluminium and, more often, loss of ground clearance through sag. It is the only one of the three that is a property of the equipment rather than of the system, and it is weather- and time-dependent, which is what dynamic line rating exploits.

Voltage drop. The receiving-end voltage has to stay inside its band under load. Written on the two-port,

$$U_S = |\underline{A} \cdot \underline{U}_R + \underline{B} \cdot \underline{I}_R| \leq 1.0\,\text{p.u.} \quad \text{with} \quad U_R \geq 0.95\,\text{p.u.} \tag{26}$$

Beyond natural loading the line is a net consumer of reactive power, and the drop grows faster than the transfer. Shunt compensation at the receiving end raises this limit at the cost of a weaker voltage-stability margin.

Steady-state stability. The transmission angle must stay far enough below the crest of the power-angle curve that the system survives the loss of a parallel element. With a midpoint series bank of degree k the lossless form is

$$P_{stab} \approx \frac{U^2}{(1-k) \cdot X_L} \cdot \sin \delta_{max} \tag{27}$$

and a criterion of $\delta_{max} = 30°$ leaves a factor two on the lossless maximum. Section 4.7 treats this limit properly, including why the usable value is well below P_{max}.

The loadability of the line is the minimum of the three. Plotted against length it is the St. Clair curve.

4.6.2 Loadability of the reference line

Figure 12 evaluates the three criteria on the 500 kV reference line of Section 4.4 with the exact two-port relations, for two boundary conditions. In case (a) the load sits at the receiving end with no reactive support, which is the classical St. Clair configuration. In case (b) the line runs between two stiff systems with both terminals held at 1.0 p.u., which removes the voltage criterion entirely. The thermal limit is $I_{th} = 3 \times 1300$ A at 75 °C, giving $P_{th} = 3377$ MW against $P_{nat} = 939$ MW.

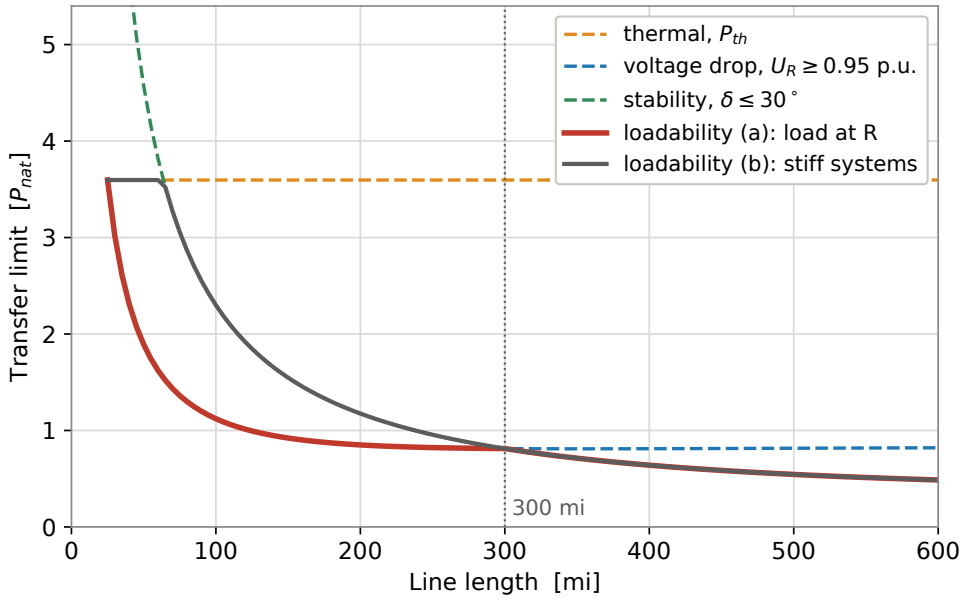


Figure 12: Loadability of the 500 kV reference line against length, cases (a) and (b), uncompensated

In case (a) the thermal limit binds only to about 25 mi, after which voltage drop takes over, and the loadability falls through $1 \cdot P_{nat}$ at 125 mi. In case (b) the thermal limit holds to about 60 mi and angular stability binds beyond it, with the crossing at 240 mi. At the 300 mi design length both cases give 763 MW, or $0.81 \cdot P_{nat}$ — the voltage and the angle criterion happen to coincide there, and the line cannot carry even its natural power without compensation.

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The boundary condition is not a detail. Holding the receiving end at 1.0 p.u. moves the $1 \cdot P_{nat}$ crossing by 115 mi on this line. Any loadability figure has to be quoted with the reactive support it assumes.

4.6.3 The same analysis on the 400 kV line

Figure 13 repeats the exercise on the 400 kV, 50 Hz line of Section 4.5, with criteria $I \le 2.0$ kA, $U_R \ge 0.95$ p.u. and $\vartheta \le 30°$. In case (a) the thermal limit holds to 80 km at $2.3 \ldots 2.4 \cdot P_{nat}$, voltage drop binds from 120 to 400 km, angular stability beyond 500 km, and the curve falls through $1 \cdot P_{nat}$ at 382 km. In case (b) the thermal limit holds to 160 km, stability binds beyond 200 km, and the crossing moves to 472 km.

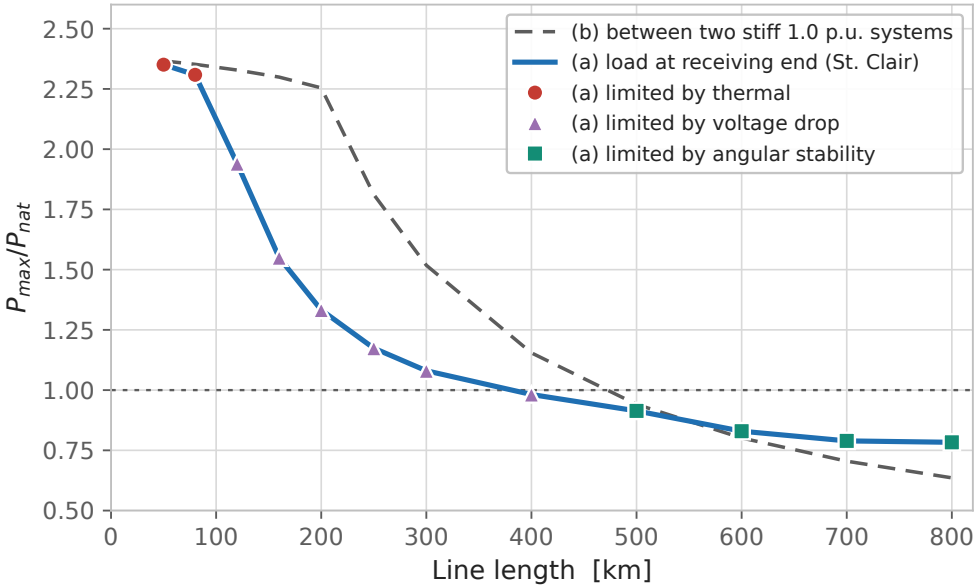


Figure 13: Loadability of the 400 kV line, showing that natural power is a reference and not a limit

The shape is the same on both lines: a thermal plateau of roughly $2 \ldots 2.5 \cdot P_{nat}$ over the first tens of kilometres, then a decay that crosses natural power somewhere between a quarter and a half of a thousand kilometres. That is the general statement the St. Clair curve makes.

4.6.4 What compensation buys

Figure 14 repeats case (b) for a midpoint series bank of degree k, and Table 14 gives the values at the 300 mi design length.

k	P_{stab}	P_U	case (b)	thermal to
	MW	MW	P_{nat}	mi
0	763	763	0.81	60
0.3	1027	826	1.09	85
0.5	1311	844	1.40	110
0.7	1708	796	1.82	145

Table 14: Effect of a midpoint series bank at 300 mi on the reference line

The bank acts on the stability limit: $k = 0.5$ lifts the 300 mi loadability of case (b) from 0.81 to $1.40 \cdot P_{nat}$. The voltage limit of case (a) barely moves, because the $I \cdot X_C$ rise across the bank is in quadrature with a unity-power-factor load. Series compensation therefore does not remove the need for reactive support at the receiving end; it addresses a different limit.

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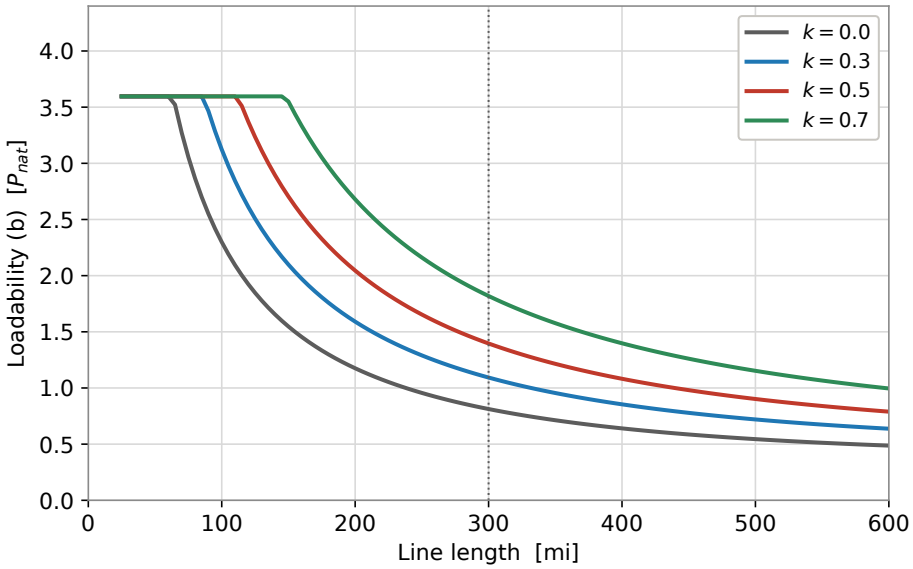


Figure 14: Case (b) loadability of the reference line with a midpoint series bank of degree k

4.6.5 Equipment limits

The conductor is not always the first thing to run out. Currents follow from the line-to-line voltage,

$$I = \frac{S}{\sqrt{3} \cdot U_{LL}} \tag{28}$$

and the switchgear in series with the line carries the same current. The largest continuous rating available in transmission circuit breakers is about 5000 A (Hitachi Energy, GE), so on a high-current scheme the breaker rating, not the conductor, can set the maximum current and force either a higher voltage or a second circuit.

4.7 Angular Stability

Section 4.6 introduced the angular criterion as one of three bounds. On any line long enough to matter it is the one that binds, and the margin it needs is larger than the criterion alone suggests.

4.7.1 Synchronising power

The meaningful quantity is not the crest of the power-angle curve but its slope, the *synchronising power coefficient*

$$K_s = \frac{\partial P}{\partial \delta} = \frac{U_S \cdot U_R}{X} \cdot \cos \delta \tag{29}$$

which is the stiffness of the electrical coupling between the machines at either end. Below the maximum, a rotor that advances delivers more power, decelerates and returns — negative feedback. Beyond it, advancing delivers less power and the machine accelerates further; $K_s = 0$ at the maximum is therefore the steady-state stability limit in the strict sense. Figure 15 shows both curves for the reference line.

With losses the maximum does not fall at $90°$ but at $\delta = \vartheta_B$, which is $81.0°$ on the reference line. At $\delta = 30°$ the line carries $0.81 \cdot P_{nat}$ with 79 % of the stiffness it has at $\delta = 0$.

Linearising the swing equation about an operating point gives

$$\frac{2 \cdot H}{\omega_0} \cdot \Delta\ddot{\delta} + K_D \cdot \Delta\dot{\delta} + K_s \cdot \Delta\delta = 0 \qquad \omega_n = \sqrt{\frac{K_s \cdot \omega_0}{2 \cdot H}} \tag{30}$$

with H the inertia constant and K_D the damping torque coefficient. As δ grows, K_s falls, the electromechanical mode slows and its damping ratio drops. The machine becomes sluggish and poorly damped well before it actually pulls out, which is the real reason nobody operates near the maximum.

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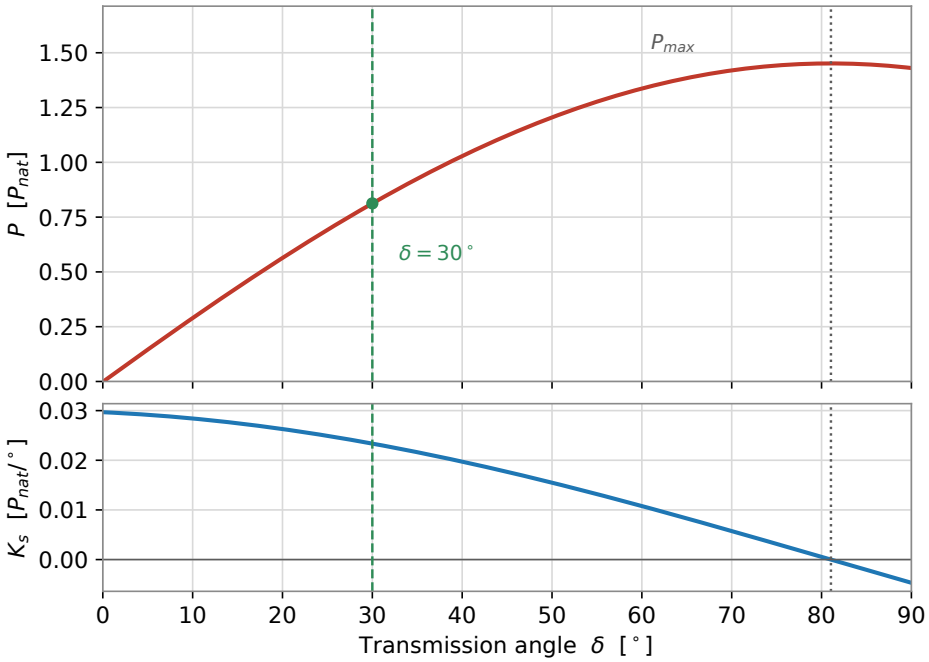


Figure 15: Power-angle relation and synchronising power coefficient of the 500 kV reference line at 300 mi, both terminals at 1.0 p.u.

4.7.2 Electrical length

For anything over roughly 250 km the lumped $\omega \cdot L' \cdot l$ overstates the reactance. The series element of the exact equivalent is $X' = Z_w \cdot \sin(\beta \cdot l)$, and substituting into the lossless maximum gives a clean result:

$$P_{max} = \frac{U^2}{Z_w \cdot \sin(\beta \cdot l)} = \frac{P_{nat}}{\sin(\beta \cdot l)} \tag{31}$$

At a quarter wavelength — 1234 km on the reference line at 60 Hz — the maximum transfer is exactly P_{nat}. That is the analytical spine beneath the St. Clair curve, and the reason capability decays with length rather than staying flat. Figure 16 compares the three against length.

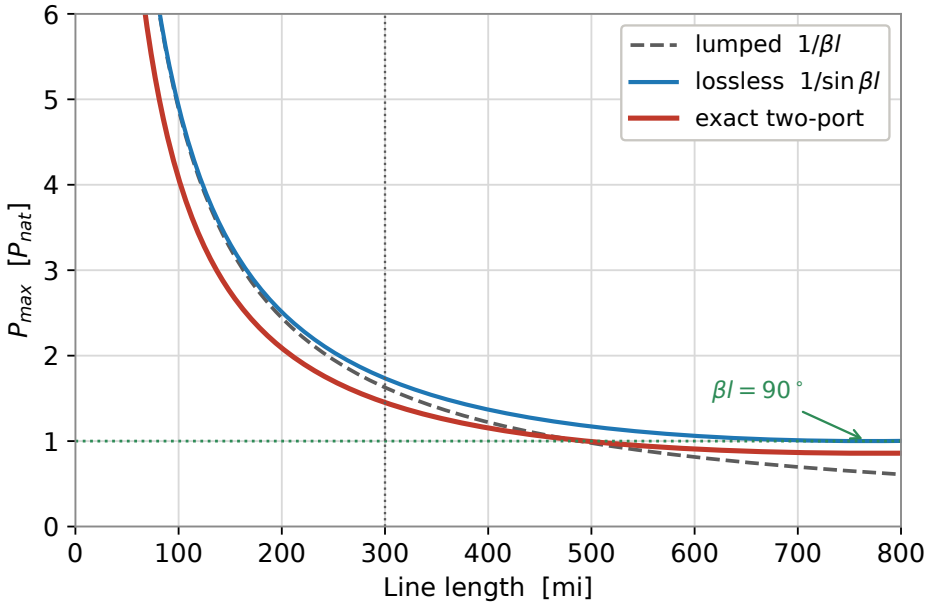


Figure 16: Maximum transfer against length: lumped, lossless distributed and exact two-port

At the 300 mi design length $\beta \cdot l = 35.2°$, the lumped reactance is $162.4\,\Omega$ against $X' = 153.4\,\Omega$, and

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$1/\sin(\beta \cdot l) = 1.73$ against 1.45 for the exact two-port; losses and charging account for the remaining gap.

4.7.3 From the maximum to the operating limit

Two effects cut the usable transfer well below P_{max}, and they compound.

The first is the post-contingency network. Two parallel 300 mi circuits have $P_{max} = 2726$ MW intact; losing one doubles the reactance of the path and leaves 1363 MW on the survivor. Planning practice requires a margin of 30 to 35 % on the *post-contingency* network,

$$\frac{P_{max} - P}{P_{max}} \geq 0.30 \tag{32}$$

which on this pair allows 954 MW, or $1.02 \cdot P_{nat}$ for the two circuits together. A transfer of 1200 MW looks comfortable intact, at $\delta = 22.9°$, and sits only 12 % below the post-contingency maximum — not acceptable. For comparison, the thermal rating of the pair is 6755 MW, seven times the binding constraint. Figure 17 shows both power-angle curves.

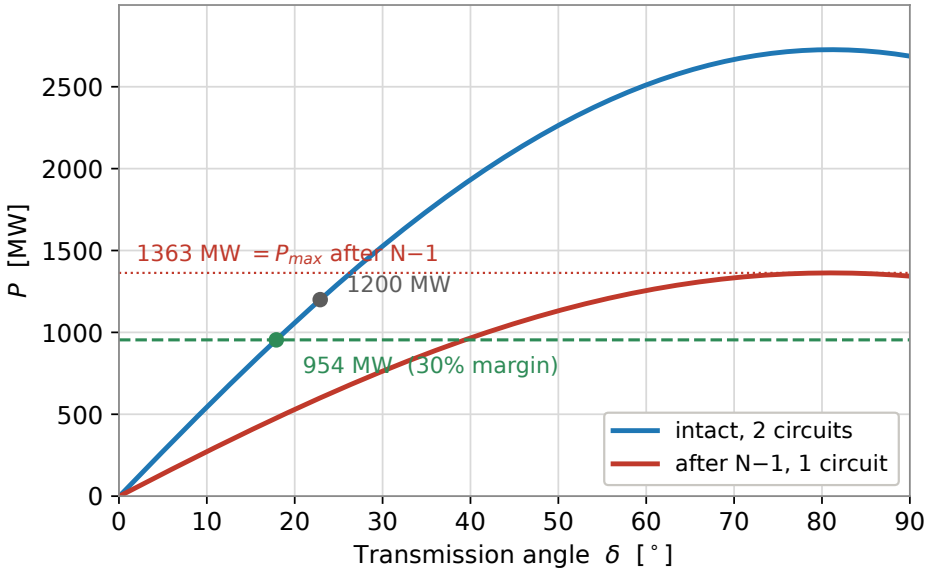


Figure 17: Power-angle curves of two parallel 300 mi circuits, intact and after the loss of one

4.7.4 Transient stability

The second effect is the fault case. During a fault the electrical output collapses while the mechanical input continues; the rotor must give that energy back before the maximum angle is reached. The equal-area criterion states the condition as

$$A_{acc} = \int_{\delta_0}^{\delta_{cc}} (P_m - P_{flt})\,\mathrm{d}\delta \;=\; A_{dec} = \int_{\delta_{cc}}^{\delta_{max}} (P_{pst} - P_m)\,\mathrm{d}\delta \tag{33}$$

with δ_{cc} the clearing angle. For a machine at $P_m = 0.6$ p.u. with maxima of 1.0 pre-fault, 0.3 during the fault and 0.7 after it, $\delta_0 = 36.9°$, $\delta_{cc} = 45.5°$ and $\delta_{max} = 121.0°$, Figure 18.

A higher pre-fault angle shrinks the decelerating area from both ends at once, so the critical clearing time falls steeply with loading. This is why two-cycle breakers and independent-pole tripping buy transfer capability, and why the transient limit usually binds before the steady-state one.

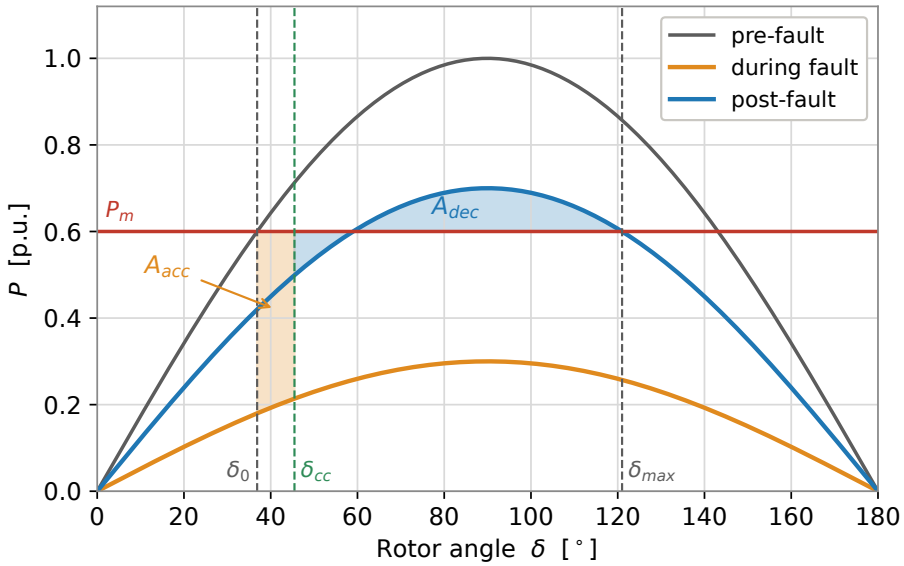


Figure 18: Equal-area criterion, generic machine per unit

4.7.5 Damping

In a multi-machine system the classical limit generalises to an eigenvalue criterion. The modes that constrain long transfers are the inter-area modes at 0.2 to 0.8 Hz; local plant modes sit at 0.8 to 2 Hz and torsional modes far above. The practical limit is where the damping ratio of a critical mode falls below about 3 to 5 %, which can occur at angles well under $40°$ and therefore well below the two-machine maximum.

High-gain automatic voltage regulators, which are what allow a large angle in the first place, contribute *negative* damping torque at heavy loading. The power system stabiliser exists to put it back. Tuning it, not raising P_{max}, is often what releases transfer on a long corridor.

4.7.6 Why a DC link has none of this

A DC link has no transmission angle, no synchronising torque and no equal-area criterion. The converter control sets power as a commanded quantity, and length enters only through the conductor resistance, so the $1/\sin(\beta \cdot l)$ decay of (31) has no counterpart. An embedded DC link can be modulated to damp the inter-area modes of the surrounding AC system instead of loading them.

The constraint moves to the terminals: short-circuit ratio and commutation robustness for line-commutated converters, control interaction for voltage-sourced ones. That is the trade the rest of this report is about.

4.8 Series Compensation

4.8.1 Principle

A capacitor bank in series with the line cancels part of its inductive reactance. The degree of compensation is defined against the total series reactance,

$$k = \frac{X_C}{X_L}, \qquad X_L = \omega \cdot L' \cdot l, \qquad X_{eff} = (1 - k) \cdot X_L \tag{34}$$

so that the lossless transfer becomes

$$P = \frac{U^2}{(1 - k) \cdot X_L} \cdot \sin\delta \tag{35}$$

with $|U_S| = |U_R| = U$. The line behaves as though it were shorter: for a given transfer the angle falls and the stability margin of Section 4.7 rises.

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Two properties distinguish it from shunt compensation. The voltage rise $I \cdot X_C$ across the bank scales with the current, so the compensation is self-regulating: it appears when it is needed and not at light load. And it acts on the series branch, which is why it addresses the angular limit and barely moves the voltage limit of case (a) in Section 4.6.4.

The practical range is $k \approx 0.25 \ldots 0.70$. Above that, sub-synchronous resonance, distance-protection behaviour and bank cost dominate the choice.

In the two-port model the bank is cascaded as a series element at the point where it is installed,

$$\begin{bmatrix} 1 & -j \cdot X_C \\ 0 & 1 \end{bmatrix} \tag{36}$$

between the two half-lines for a midpoint bank, which is the arrangement used throughout this section.

4.8.2 Single-line arrangement

A series bank is not only capacitors. The platform sits at line potential and carries the full fault current until it is bypassed, so each phase carries a metal-oxide varistor across the bank to limit the capacitor voltage under fault current, a bypass breaker with a damping reactor to take over when the varistor reaches its energy limit, and disconnectors to isolate the platform. A bypass disconnector keeps the line in service with the bank out, at $k = 0$. Figure 19 shows the arrangement for one phase, with the bank at the midpoint of the line.

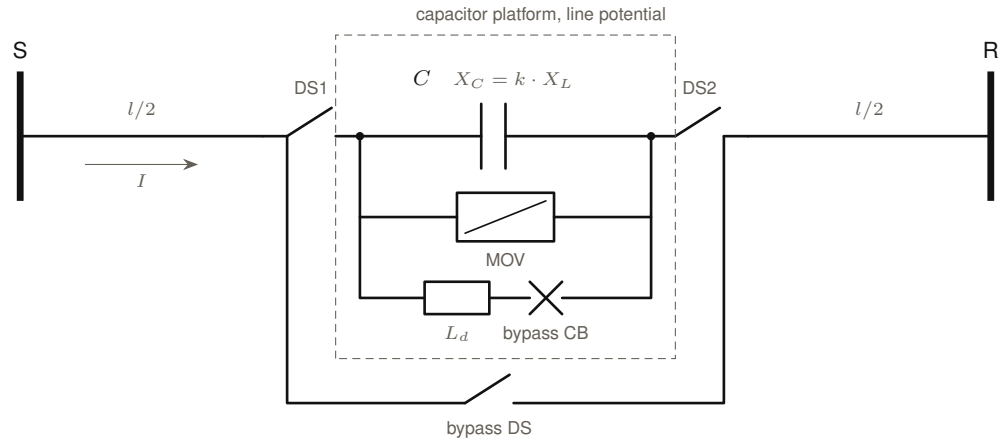


Figure 19: Single-line diagram of a midpoint series bank, one phase. Symbols to IEC 60617 [18]

The varistor limits the capacitor voltage under fault current and the bypass breaker with its damping reactor L_d takes over at the varistor energy limit. DS1 and DS2 isolate the platform; the bypass disconnector keeps the line in service with the bank out.

4.8.3 Effect on the power-angle curve

Figure 20 shows the power-angle curves of the 500 kV reference line for a midpoint bank at several degrees of compensation, and Table 15 the resulting maxima.

Uncompensated, 1500 MW — $1.6 \cdot P_{nat}$ — cannot be carried at all; at $k = 0.5$ it is carried at $36°$. The maximum also moves to smaller angles as k grows, which is the electrical shortening of the line expressed on the curve. The lossless form $U^2 / ((1 - k) \cdot X_L)$ would give 1540 MW uncompensated and 3080 MW at $k = 0.5$; at this length $R = 27.5\,\Omega$ and the line charging are not negligible, which is why the exact two-port is used.

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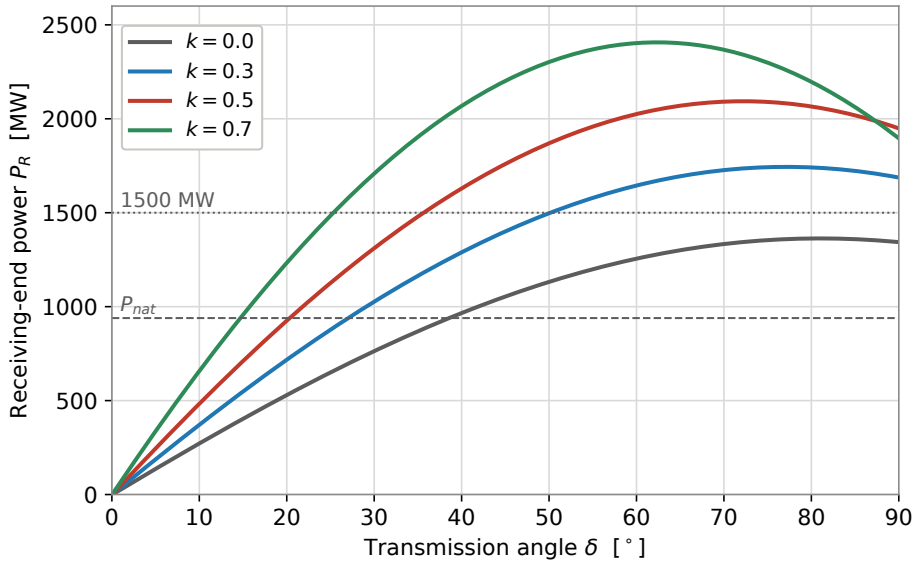


Figure 20: Power-angle curves of the reference line with a midpoint series bank, both terminals at 1.0 p.u.

k	X_C	P_{max}	δ **at 1500 MW**
	Ω	MW	°
0	0	1363	—
0.3	48.7	1744	50.1
0.5	81.2	2093	35.7
0.7	113.7	2407	25.3

Table 15: Midpoint bank on the 300 mi reference line, exact two-port

4.8.4 Voltage profile at heavy load

Delivering 900 MW at $\cos\varphi = 0.95$ lagging with the receiving end held at 1.0 p.u., the uncompensated line requires a sending-end voltage of 1.217 p.u., far outside any operating band. The midpoint bank lifts the voltage by $I \cdot X_C$ in the direction of power flow, and the sending end falls to 1.157, 1.127 and 1.107 p.u. for $k = 0.3$, 0.5 and 0.7, Figure 21.

The receiving half of the line is unchanged, because the bank acts only on what is upstream of it. That is why end-of-line banks and split banks are also used where the voltage profile rather than the angle is the problem.

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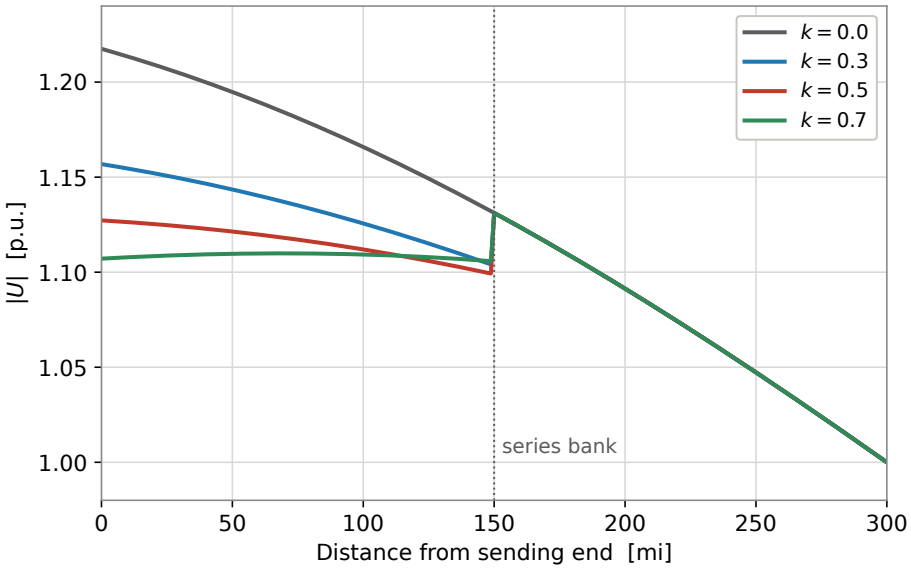


Figure 21: Voltage profile at heavy load with a midpoint series bank

4.8.5 Bank rating and protection

The bank is rated for the planned continuous line current, not for the conductor thermal limit:

$$U_C = I_{rated} \cdot X_C \qquad Q_C = 3 \cdot I_{rated}^2 \cdot X_C \tag{37}$$

Table 16 gives the resulting bank for the reference line at $k = 0.5$.

Quantity		Value
k		0.5
X_C	Ω	81.2
C per phase	µF	32.7
I_{rated}	A	2000
U_C per phase	kV	162
Q_C three-phase	MVAr	974
MOV level ($2.3 \cdot U_C$)	kV	373

Table 16: Midpoint bank rating on the reference line at $k = 0.5$

I_{rated} is taken as about $1.15\cdot$ the current at 1500 MW and 500 kV. The fault-current voltage is limited by the metal-oxide varistor at a protective level of $2.0 \dots 2.5 \cdot U_C$, with the bypass breaker and damping circuit behind it. Emergency ratings — typically 30 min and 10 s — are specified separately from the continuous rating.

4.8.6 Sub-synchronous resonance

The bank forms a series resonance with the line and source reactances at

$$f_{er} = f_0 \cdot \sqrt{\frac{X_C}{X_L + X_S}} \tag{38}$$

Rotor currents see the complement $f_0 - f_{er}$, which has to stay clear of the torsional modes of the turbine-generator shaft — 10 to 45 Hz for large steam units. Figure 22 screens the reference line against the degree of compensation.

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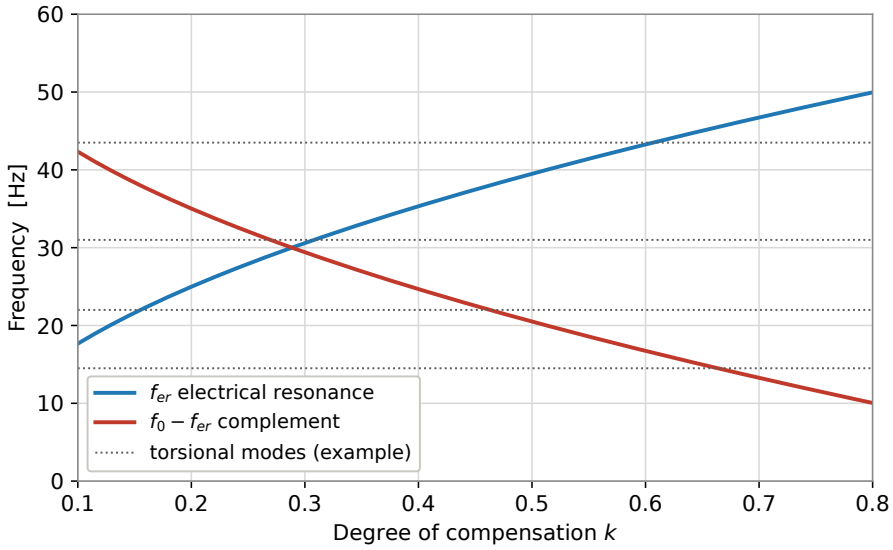


Figure 22: Sub-synchronous resonance screening: electrical resonance and its complement against the degree of compensation

With $X_S = 25\,\Omega$, which is 15 % on 1500 MVA, a compensation of $k = 0.5$ gives $f_{er} = 39.5$ Hz and a complement of 20.5 Hz — within 2 Hz of an illustrative 22 Hz torsional mode. A hit at screening level calls for a detailed study or a different k; type-3 wind generation adds sub-synchronous control interaction to the same screening. The torsional modes used at this stage are illustrative, and real values come from the generator manufacturer.

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A Two-Port Networks

This appendix is background. Section 4 is self-contained and nothing here is needed to follow it; what follows is for the reader who wants to know where its equations come from and how far they reach.

The chain of matrices that Section 4 uses to describe a transmission line is not a power-transmission construction. It is the standard description of any linear network reached through two terminal pairs, and it was in use for telephone repeaters and filters before it was applied to power lines. Setting it out on its own terms shows the line to be one instance of it rather than a special theory, and shows which of its results are properties of a line and which are properties of any two-port at all.

Two properties make the description worth the abstraction. The first is that it hides everything inside the network: two ports and four complex numbers describe the terminal behaviour completely, whatever the internal arrangement. The second is that networks in cascade multiply, so a chain of elements of any length collapses to one matrix of the same size.

A.1 The port concept

A *port* is a pair of terminals through which the current entering one terminal equals the current leaving the other. That equality is the port condition, and it is an assumption about how the network is connected, not a property of the network itself: it fails if a third path — a stray capacitance to earth, a shared return, a ground loop — carries current between the two sides. A two-port is a network with exactly two such pairs, Figure A.1.

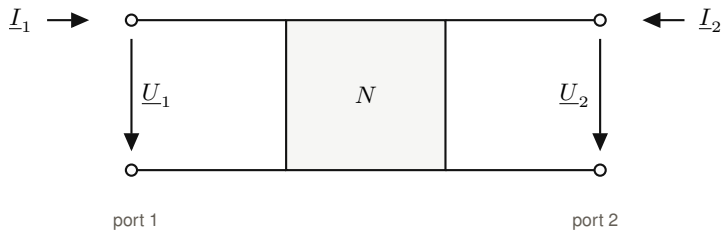


Figure A.1: A two-port and its terminal quantities, drawn in the convention in which both port currents enter the network

Four quantities appear at the terminals: $\underline{U}_1$, $\underline{I}_1$, $\underline{U}_2$ and $\underline{I}_2$. The network imposes two linear relations between them, so any two may be chosen as independent and the other two follow. Which two are chosen is the only difference between the parameter sets below.

A.2 Parameter sets

Table A.1 lists the six choices. The impedance, admittance and the two hybrid sets are drawn in the convention of Figure A.1, with both currents entering the network. The chain sets reverse the reference direction at port 2, so that $\underline{I}_2$ leaves the network towards whatever is connected next — without that reversal, cascading would alternate in sign at every junction. Comparing results from two sources that differ only in this convention is the most common error in using them.

The sets are equivalent wherever they all exist, and each is undefined for some networks: an ideal transformer has no impedance matrix, an ideal series element no admittance matrix. The chain matrix exists unless the network cannot transfer anything between the ports at all.

A.3 Chain parameters and cascading

The chain form writes port 1 in terms of port 2,

$$\begin{bmatrix} \underline{U}_1 \\ \underline{I}_1 \end{bmatrix} = \underline{T} \cdot \begin{bmatrix} \underline{U}_2 \\ \underline{I}_2 \end{bmatrix}, \qquad \underline{T} = \begin{bmatrix} \underline{A} & \underline{B} \\ \underline{C} & \underline{D} \end{bmatrix} \tag{A.1}$$

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Set	Relation	Typical use
Impedance $\underline{Z}$	$\underline{U} = \underline{Z} \cdot \underline{I}$	Networks joined in series; open-circuit measurements
Admittance $\underline{Y}$	$\underline{I} = \underline{Y} \cdot \underline{U}$	Networks joined in parallel; nodal analysis
Hybrid $\underline{h}$	$(\underline{U}_1, \underline{I}_2)$ from $(\underline{I}_1, \underline{U}_2)$	Amplifying devices, where the input is a current and the output a voltage
Inverse hybrid $\underline{g}$	$(\underline{I}_1, \underline{U}_2)$ from $(\underline{U}_1, \underline{I}_2)$	The dual of the above
Chain $\underline{T}$ (ABCD)	port 1 from port 2	Elements in cascade; transmission paths
Inverse chain	port 2 from port 1	The same, traversed in the other direction

Table A.1: The six parameter sets of a two-port, distinguished only by which two terminal quantities are taken as independent [19], [20]

with $\underline{I}_2$ leaving the network. The four entries have a meaning one at a time. With port 2 open, $\underline{I}_2 = 0$ and $\underline{A} = \underline{U}_1/\underline{U}_2$ is the reciprocal voltage ratio, while $\underline{C} = \underline{I}_1/\underline{U}_2$ is a transfer admittance. With port 2 shorted, $\underline{U}_2 = 0$ and $\underline{B} = \underline{U}_1/\underline{I}_2$ is a transfer impedance, $\underline{D} = \underline{I}_1/\underline{I}_2$ the reciprocal current ratio. So $\underline{A}$ and $\underline{D}$ are dimensionless, $\underline{B}$ is an impedance and $\underline{C}$ an admittance — which is why the four cannot be compared with each other as magnitudes.

The reason this set is the one used for transmission is cascading. If two two-ports are connected so that port 2 of the first is port 1 of the second, Figure A.2, the intermediate quantities are shared and eliminate, leaving

$$\underline{T} = \underline{T}_a \cdot \underline{T}_b, \qquad \text{and in general} \qquad \underline{T} = \prod_{i=1}^{n} \underline{T}_i \tag{A.2}$$

in the order the elements are traversed. The product does not commute: reversing two elements of a chain generally changes the result, which is the algebraic statement that it matters where in a path an element sits. A compensating element at the middle of a line and the same element at its end are different networks, and (A.2) is what distinguishes them.

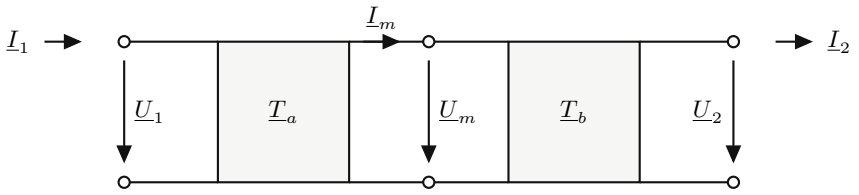


Figure A.2: Two two-ports in cascade: the intermediate quantities are common to both and eliminate, so the chain matrices multiply

A.4 Other interconnections

Cascade is one of five ways two two-ports can be joined, and each has its own parameter set in which the combination is a simple sum, Table A.2. The pattern is the same throughout: pick the set whose independent variables are the ones the connection forces to be shared, and the matrices add.

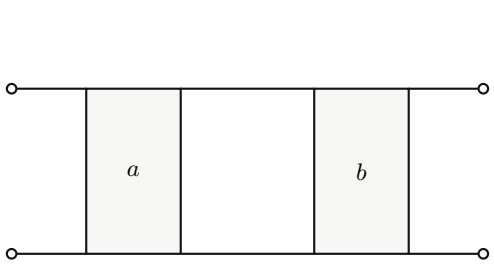
Cascade is the odd one out, and the difference is worth stating plainly because the word “series” is used loosely for both. In a cascade the networks are passed through one after another and the matrices *multiply*; in a series connection they sit side by side with their ports stacked, both carry the same currents, and the matrices *add*, Figure A.3. A capacitor bank inserted in a line is a cascade, not a series connection of two-ports, even though the capacitor is in series with the conductor: what is in series there is a circuit element within one port, not two ports with each other.

The addition rules carry a condition that the cascade does not. Stacking two networks can destroy the port condition inside them: current that entered one terminal of a sub-network may return through the other sub-network rather than through its partner terminal, and then the sub-network no longer has the matrix it was given alone, so the sum is wrong. Whether a particular pairing is safe is settled by a test due to Brune,

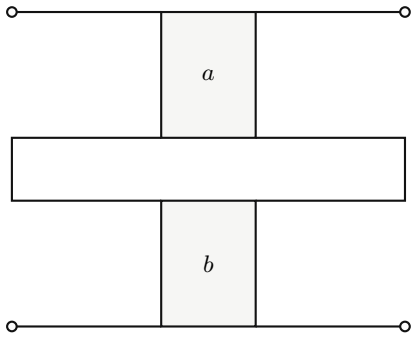
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Connection	Shared	Rule	Where it is met
Cascade	port 2 of one is port 1 of the next	$\underline{T} = \underline{T}_a \cdot \underline{T}_b$	Line sections, a series bank between two half-lines, a transformer ahead of a line
Series	both currents	$\underline{Z} = \underline{Z}_a + \underline{Z}_b$	Windings in series, impedances in a common current path
Parallel	both voltages	$\underline{Y} = \underline{Y}_a + \underline{Y}_b$	Parallel circuits of a double-circuit line, transformers in parallel
Series-parallel	$\underline{I}_1$ and $\underline{U}_2$	$\underline{h} = \underline{h}_a + \underline{h}_b$	Feedback arrangements in amplifiers
Parallel-series	$\underline{U}_1$ and $\underline{I}_2$	$\underline{g} = \underline{g}_a + \underline{g}_b$	The dual of the above

Table A.2: The five interconnections of two two-ports and the parameter set in which each is additive [19], [20]



(a) cascade: $\underline{T} = \underline{T}_a \cdot \underline{T}_b$



(b) series: $\underline{Z} = \underline{Z}_a + \underline{Z}_b$

Figure A.3: Cascade against series connection. In the cascade the networks are passed through one after the other and the chain matrices multiply; in the series connection the ports are stacked, both networks carry the same currents, and the impedance matrices add

applied to the two networks as connected [20]. Two cases are safe without testing: either sub-network having an isolating transformer at one port, and — in power work the common case — two networks that share a common return and are each symmetric about it. The cascade needs no such test, because it adds no new connection between terminals that were not already a port.

The parallel rule is the one met most often in transmission. Two identical circuits on the same route, each with parameters $\underline{A} \ldots \underline{D}$, give

$$\underline{A}_{\parallel} = \underline{A}, \qquad \underline{B}_{\parallel} = \frac{\underline{B}}{2}, \qquad \underline{C}_{\parallel} = 2 \cdot \underline{C}, \qquad \underline{D}_{\parallel} = \underline{D} \tag{A.3}$$

which is the statement that two circuits halve the transfer impedance and so double the maximum transfer, while the voltage ratio is unchanged. Losing one of them restores $\underline{B}$, which is the contingency case of Section 4.7.3. Equation (A.3) follows from the admittance rule, and is quicker to use than converting twice.

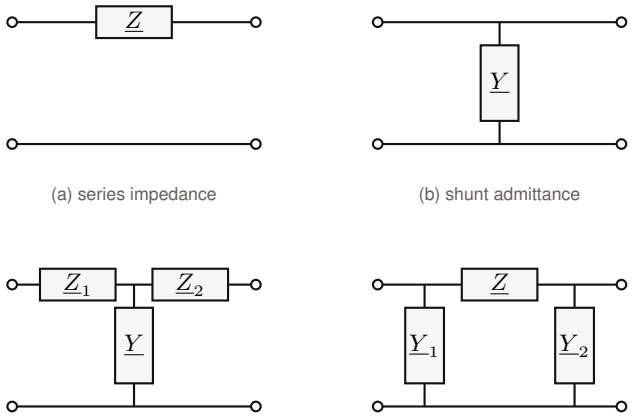
A.5 Elementary two-ports

Every ladder network is built from two elements: an impedance in series with one rail and an admittance across the rails, Figure A.4(a) and (b). Their chain matrices are

$$\underline{T}_{series} = \begin{bmatrix} 1 & \underline{Z} \\ 0 & 1 \end{bmatrix}, \qquad \underline{T}_{shunt} = \begin{bmatrix} 1 & 0 \\ \underline{Y} & 1 \end{bmatrix} \tag{A.4}$$

and everything else in Table A.3 is a product of these two, taken in the order the elements are met.

The last row is the one that matters for a transmission line and is derived in Section 4.1.5; it is included here to show that it is the same object as the others, not a separate theory.



(a) series impedance (b) shunt admittance

(c) T-section (d) π-section

Figure A.4: The two elementary two-ports and the two ladder sections built from them

Network	$\underline{A} \dots \underline{D}$	**Remark**
Series $\underline{Z}$	$1,\ \underline{Z},\ 0,\ 1$	$\underline{A} = 1$: no shunt path, so the open-circuit voltage ratio is unity
Shunt $\underline{Y}$	$1,\ 0,\ \underline{Y},\ 1$	$\underline{B} = 0$: the ports are directly connected
Symmetric π	$1 + \frac{\underline{Z}\cdot\underline{Y}}{2},\ \underline{Z},\ \underline{Y}\cdot\left(1 + \frac{\underline{Z}\cdot\underline{Y}}{4}\right),\ 1 + \frac{\underline{Z}\cdot\underline{Y}}{2}$	$\underline{Y}/2$ at each end; the nominal line section of (4)
Ideal transformer	$n,\ 0,\ 0,\ 1/n$	$\underline{B} = \underline{C} = 0$: a pure scaling of both quantities
Uniform line	$\cosh\underline{\theta},\ \underline{Z}_w \cdot \sinh\underline{\theta},\ \sinh\underline{\theta}/\underline{Z}_w,\ \cosh\underline{\theta}$	$\underline{\theta} = \underline{\gamma}\cdot l$; the limit of infinitely many π-sections

Table A.3: Chain matrices of the elementary two-ports and of the uniform line [19], [21]

A.6 Reciprocity, symmetry and losslessness

Three structural properties of a network show themselves directly in the four entries.

Reciprocity. A network built only from impedances, whatever their arrangement, satisfies

$$\underline{A}\cdot\underline{D} - \underline{B}\cdot\underline{C} = 1 \tag{A.5}$$

Physically: a source at port 1 and an ammeter at port 2 read the same as the source and the ammeter exchanged. It fails for networks containing an independent source or a non-reciprocal element such as a gyrator or a magnetised ferrite. Because the determinant is unity, the product in (A.2) preserves it, so any cascade of reciprocal networks is reciprocal.

Symmetry. If the network looks the same from both ports then

$$\underline{A} = \underline{D} \tag{A.6}$$

Every uniform line is symmetric; a π-section with unequal shunt branches is not, and neither is a bank placed anywhere other than at the midpoint. Symmetry is the condition under which the image impedance of Section A.7 is the same at both ends.

Losslessness. A network of pure reactances has $\underline{A}$ and $\underline{D}$ real and $\underline{B}$ and $\underline{C}$ purely imaginary. The departure of $\underline{A}$ from a real number is therefore a direct measure of the loss in a line, and is why the lossless approximations of Section 4 can be stated by dropping real parts rather than by rebuilding the model.

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A.7 Terminating a two-port

A two-port on its own is not a circuit. Closing port 2 with an impedance $\underline{Z}_L$ makes $\underline{U}_2 = \underline{Z}_L \cdot \underline{I}_2$, and (A.1) gives the impedance the source at port 1 sees:

$$\underline{Z}_{in} = \frac{\underline{U}_1}{\underline{I}_1} = \frac{\underline{A} \cdot \underline{Z}_L + \underline{B}}{\underline{C} \cdot \underline{Z}_L + \underline{D}} \tag{A.7}$$

This is the single most used result of the whole description: every statement about what a network does to the thing behind it is a special case of it. With port 2 open it gives $\underline{A}/\underline{C}$, with port 2 shorted $\underline{B}/\underline{D}$.

One termination is distinguished. On a symmetric network there is a value of $\underline{Z}_L$ for which $\underline{Z}_{in} = \underline{Z}_L$, so that the network is invisible to its surroundings and may be repeated any number of times without changing anything. Setting $\underline{Z}_{in} = \underline{Z}_L$ in (A.7) with $\underline{A} = \underline{D}$ gives the *image* or *characteristic* impedance

$$\underline{Z}_c = \sqrt{\frac{\underline{B}}{\underline{C}}}, \qquad \underline{\theta} = \operatorname{arcosh} \underline{A} \tag{A.8}$$

with $\underline{\theta}$ the image transfer constant, the complex number by which a quantity is scaled and shifted in passing through the network. Substituting the uniform-line row of Table A.3 returns $\underline{Z}_c = \underline{Z}_w$ and $\underline{\theta} = \underline{\gamma} \cdot l$: the wave impedance of a line and its electrical length are not line-specific quantities but this general pair, evaluated for that network. The natural power of Section 4.5 is the same statement in power terms — a line closed in its own image impedance.

A.8 The same algebra in other domains

Nothing above used the fact that the quantities were electrical. The derivation needs only a pair of variables at each port whose product is power: one that is common to the elements in parallel and one that is common to the elements in series. Table A.4 lists the pairs for which the identical algebra is standard practice.

Domain	Across	Through	Where the chain form is used
Electrical	Voltage	Current	Lines, filters, matching networks, machine equivalents
Translational	Velocity	Force	Vibration isolators, drive trains, transducer chains
Rotational	Angular velocity	Torque	Shaft lines, gearboxes, torsional models of a turbine set
Acoustic	Sound pressure	Volume velocity	Ducts, horns, mufflers, loudspeaker enclosures [22]
Thermal	Temperature difference	Heat flow	Layered walls, heat sinks, cable thermal ratings

Table A.4: Domains in which the two-port algebra is used unchanged, with the pair of port variables whose product is power [20], [22]

A transducer is simply a two-port whose ports lie in two different rows of that table: a loudspeaker is an electrical port and an acoustic one, a motor an electrical port and a rotational one, a piezoelectric stack an electrical port and a translational one. Cascading them is the same matrix product, and it is how a complete electro-mechanical chain is reduced to one matrix.

Two further uses are worth naming because they are reached from a different starting point and arrive at the same matrix.

Ray and beam optics. A sequence of lenses, mirrors and free spaces is handled by a 2×2 *ray transfer matrix* acting on the ray height and its slope, multiplied in the order the light meets the elements [23]. A thin lens is $[1, 0; -1/f, 1]$ — the shunt element of (A.4) with $\underline{Y} = -1/f$ — and a free space of length d is $[1, d; 0, 1]$, the series element. The resonator condition that the beam reproduce itself after one round trip is the image-impedance condition (A.8) in another vocabulary.

High-frequency networks. Above a few hundred megahertz a terminal voltage is not measurable and the scattering parameters are used in place of the chain matrix. They are a change of variables, not a different theory: the scattering matrix is the same network expressed in incident and reflected waves against a reference impedance, and converts back to $\underline{A} \dots \underline{D}$ in closed form. Cascading in that vocabulary uses the transfer-scattering matrix, which multiplies exactly as (A.2) does.

A.9 What the description assumes

The generality has a boundary, and the assumptions are easy to leave unstated because they hold without comment in most of this report.

- **Linear and time-invariant.** The relations are linear with constant coefficients. A saturating transformer core, a switching converter or a thyristor-controlled bank is not described by one matrix; it is described by one matrix per operating point, or by a model outside this framework.
- **One frequency.** The entries are complex numbers at a stated frequency. A chain evaluated at 60 Hz says nothing about its behaviour at 20 Hz, which is why the sub-synchronous screening of Section 4.8.6 re-evaluates the same network over a range of frequencies rather than scaling one result.
- **The port condition holds.** Current entering a terminal leaves its partner. Earth-return paths and inter-circuit coupling break it, and a line with a significant zero-sequence path is a multi-port, not a two-port.
- **No internal sources.** Independent sources inside the network add a constant term that no 2×2 matrix carries. They are handled by superposition, or by moving them outside the two-port.
- **Balanced three-phase, for the power case.** The single-line two-port of Section 4 is the positive-sequence network. Unbalance requires the sequence networks separately.

Within those bounds the description is exact, and the exactness is the point: the equations of Section 4 are not an engineering approximation to a line but the general two-port result with the line's own $\underline{A} \dots \underline{D}$ substituted.

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